Instructions

Part No.: XT0223S16125

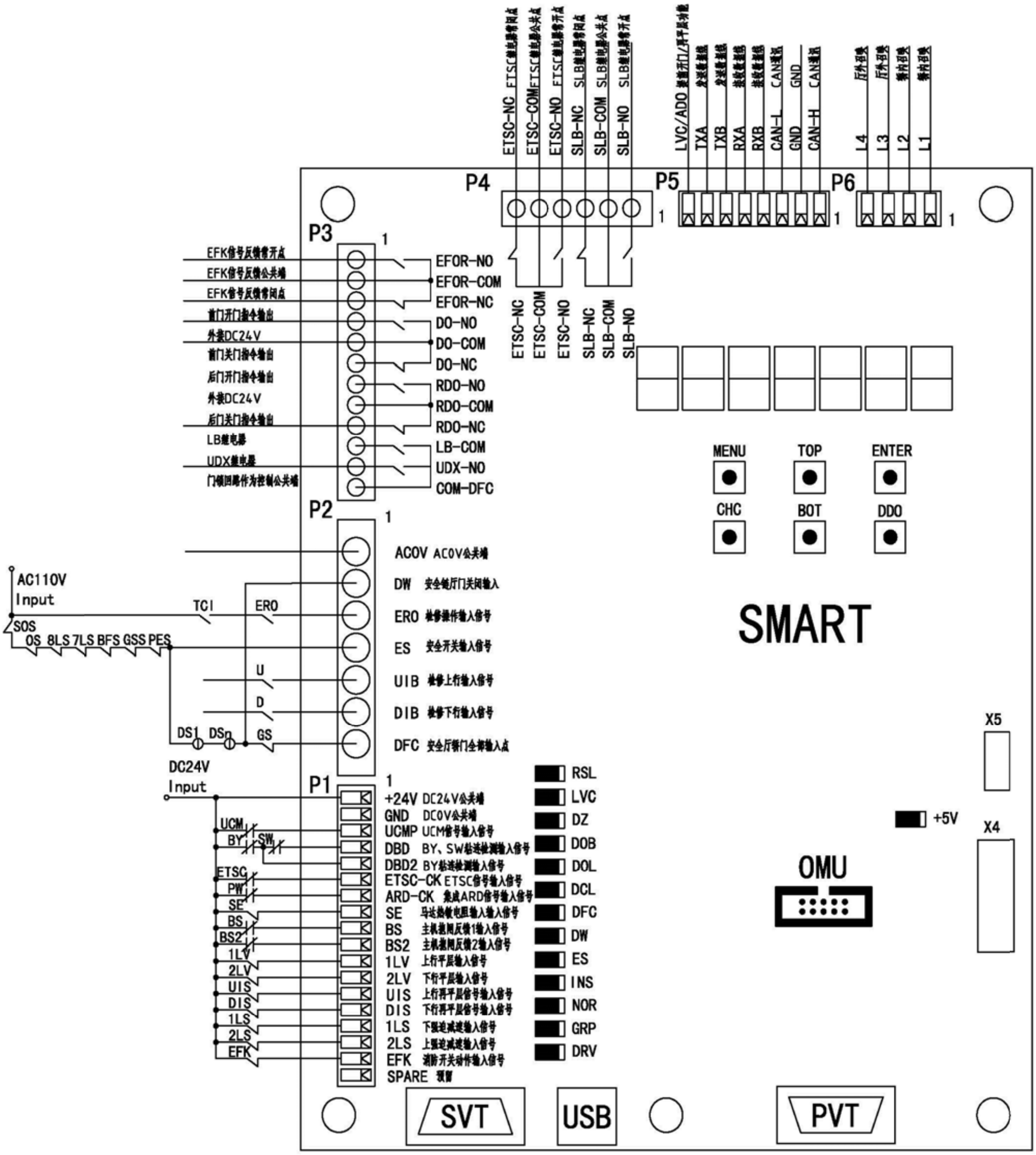
November, 2016

SMART

Instructions

1.1 Product Introduction

1.1.1 SMART External Wiring Diagram



External Wiring Diagram

1.1.2 SMART Board Indicator Specifications

Code	Function Description	Remarks
+5V	5V power normal	
RSL	Flashing indicates the communication is normal	
LVC	Light up: indicates that the bypass function is effective. Light out: indicates that the bypass function is ineffective	
DZ	Light up: indicates that the gate area signal is effective Light out: indicates that the gate area signal is ineffective	
DOB	Light up: indicates that there is a door-opening signal Light out: indicates that there is no door-opening signal	
DOL	Light up: indicates that the door-opening-in-place signal is effective Light out: indicates that the door-opening-in-place signal is ineffective	
DCL	Light up: indicates that the door-closing signal is effective Light out: indicates that the door-closing signal is ineffective	
DFC	Light up: indicates the car door is closed Light out: indicates the car door is not closed	
DW	Light up: indicates that the landing door is closed Light out: indicates that the landing door is not closed	
ES	Light up: indicates that the safety loop is disconnected Light out: indicates that the safety loop is connected	
INS	Light up: indicates it is in the state of maintenance Light out: indicates it is not in the state of maintenance	
NOR	Light up: indicates it is in a normal state Light out: indicates it is not in a normal state	
GRP	Light up: indicates it is in the group control state Light out: indicates it is not in the group control state	
DRV	Light up: indicates that the motherboard and driver communication is normal Light out: indicates that the motherboard and driver communication is not normal	

1.1.3 Serial Communication

In order to resist electromagnetic interference, it is necessary to equip the terminal of the communication line with impedance matching; the SMART board communication line is 300m in maximum length, when there is a single elevator, the impedance matching value of the terminal absorption plate inside the car and outside the hall is: 50Ω/0.47uf; when the parallel shares call or two of the groupcontrol uses the parallel display,the impedance matching value of the terminal absorption plate inside the car is: 50Ω/0.47uf and the impedance matching value of the terminal absorption plate outside the hall is 75Ω/0.33uf, and the impedance matching value of SOM board terminal absorption plate is 75Ω/0.33uf, namely SOM-75R board, shown as follows:

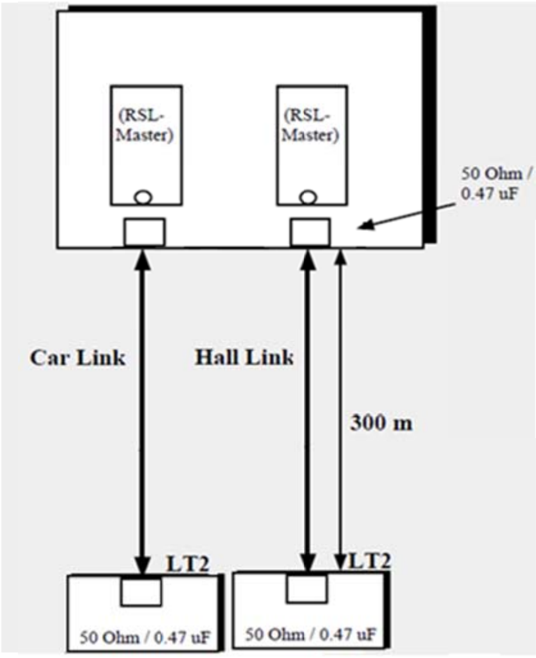


Diagram of Terminal Absorption Plate Matching of Single Elevator

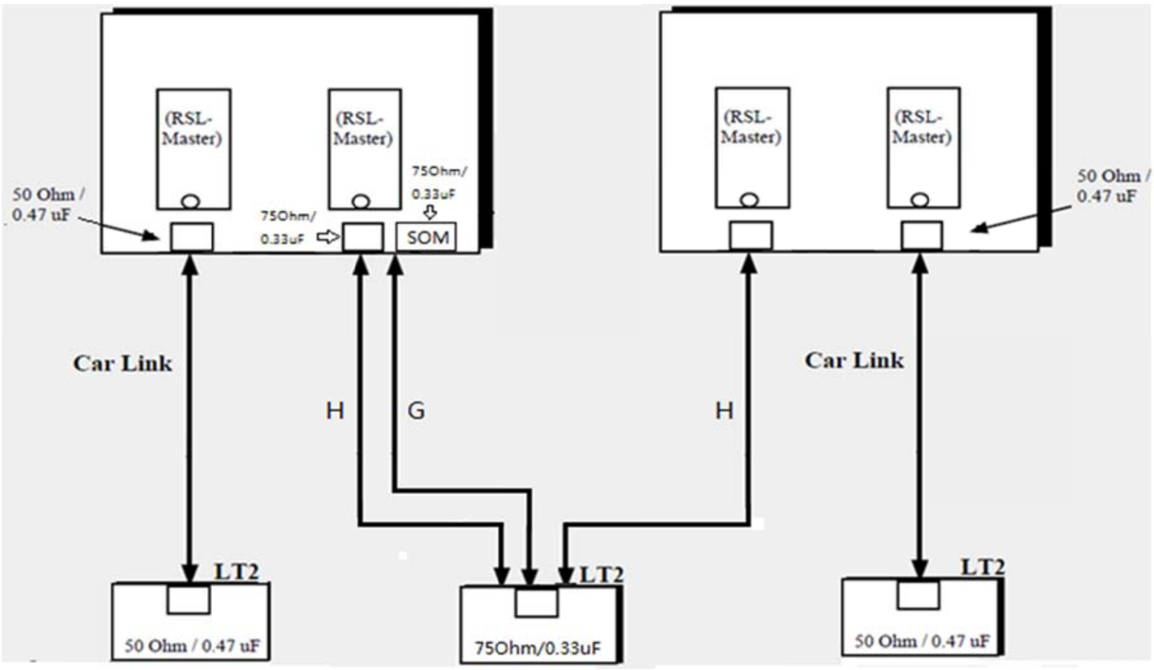


Diagram of Terminal Absorption Plate Matching of Parallel Shared Call

Note: When there is the parallel shared call or two of the group control uses the parallel display, it needs to remove the J8, J10, J15 and J16 shorted cap on SMART board.

1.1.4 Working Conditions

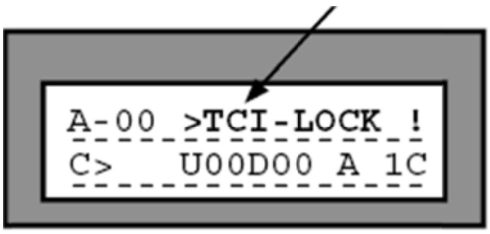
1) Not greater than 1000m above sea level;
2) Air temperature should be kept between 5~40℃;
3) The monthly average maximum relative humidity of the wettest month at the running place is 90%, while the monthly average minimum temperature of the month is not higher than 25℃;
4) Ambient air should not contain corrosive and flammable gases and conductive dust.

1.1.5 Server Monitor Interface Description

Operation Mode Description					
IDL	Normal idle mode	CHC	Shield call	ANS	Light load state
ATT	Driver service	PKS	Parking state	LNS	Full load state
EFS	Fire control state	DHB	Door holding state	OLD	Overload state
ISC	Independence state	INS	Maintenance state		
DLM	Door lock short-connection protection	HAD	Enter into the well for safety inspection		

Driving State Description					
FR	Fast running	SR	Full speed running	ST	Stop running
WT	Standby	RL	Door opening and re-leveling		

In addition, some faults will also be displayed at the location of the operating mode, such as TCI-LOCK! Flashing information;



Flashing information description is as follows:

Flashing information			
LearnRun!	Self-learning	DBSSfault	Drive not ready
DBP-Fault!	Door bypass fault	SE-Fault!	Host thermal fault
TCI-Lock!	Car top maintenance activation lock	start DCS!	Door detection sequence is not running
LS-Fault!	Forced deceleration loss	RLV-Count!	Re-leveling times out of limit
1LS+2LSon!	Up and down reduction movement	DriveFault	Drive fault
Adr-Check!	Address check	SpeedCheck	Speed feedback fault
DCS Run	Prompt door test running	DOOR bridge	Prompt door short

			connected
DC24V Lost!	24Vpower loss	HAD	Enter the well for safety inspection
DLM	Door lock short connection protection	UCM-Lock!	Car accidental movement protection

1.1.6 DCS Door Safety Inspection Function (M-1-3-5)

The function is to study and confirm the front door and rear door on each floor after the elevator well self-learning, to ensure that the front door and rear door on each floor can be normally opened and closed.

DCS Door Inspection Operation	
Button function	**Menu display**
MODULE + 1 ON + 3 DOWN + 5 SELOUT B	to start DCS press ENTER
Enter the address that needs to be searched	-00 DW:clsd <>][open front door
△▽ + CLEAR ENTER	DCS successful press GOON>
△▽ + CLEAR ENTER Continue to search	Check PES,GTC press GOON>
	to start normal press GOON>

Report Error before Operating DCS	
Error information display	**Cause of error**
DCS Start Error: Into 1LS and DZ!	Car is not on the lowest floor, move the elevator to the lowest floor
DCS Start Error: Leave 1LS!	Received the forced deceleration signal, but not received the gate area signal, move the elevator to the lowest floor
Error information display	**Cause of error**
DCS Start Error: Not able to Run!	The elevator does not run, please ensure that the elevator can be operated for maintenance
DCS Start Error: Switch off INS!	The elevator is in the maintenance mode, ensure that the elevator is in the normal mode
DCS Start Error: already done >	After completing the maintenance,press 3 DOWN button for the door safety inspection

Report Error when Operating DCS	
Error information display	**Cause of error**
Front Door Error	Front door opening error
Rear Door Error	Rear door opening error
aborted by ENTER	Server key input, DCS detection failure
DW not closed	Door is closed but the contact signal is not received
Door opening Err	Door can not be opened within 20S
Position Error	Floor number error
Door closing Err	Door can not be closed within 20S
SE is missing!	Safety loop is normal but SE signal not normal

Note: after the self learning of the well is completed, you need do DCS door state inspection. Door inspection must be done floor by floor starting from the bottom to the top. At this point, there are two ways to make the elevator run to the bottom.

Method 1: run to the bottom floor;

Method 2: M-1-3-5 menu enters to DCS interface and enter to the Sn shortcut menu of motherboard, press the motherboard BOT key, the elevator will automatically run to the bottom floor.

1.1.7 Drive Fault View Menu (M-2-3)

This menu is the drive fault view menu, press the pagedown key to view the time (need to set the standard time function) and the elevator location when the fault occurs.

M-2-3				
Display	**Serial No.**	**Value**	**Description**	**Instructions**
1) Base AD Offst E-00 Floor:01 2) 3)	**1)**	**Base AD Offst**	Error name	See error list
	2)	**E00**	Same error occurring sequence	Record 20times
	3)	**Floor:01**	Floor information	Elevator location when error occurs
Press the GO ON key				
Display	**Serial No.**	**Value**	**Description**	**Instructions**
1) Base AD Offst E-00 03-01 14:57 2) 3) 4)	1)	**Base AD Offst**	Error name	See error list
	2)	**E00**	Same error occurring sequence	Record 20times
	3)	**03-01**	Date	Date of error
	4)	**14:57**	Time	Time of error

1.1.8 Parameter Initialization Function (M-3-3)

When the parameters of the motherboard are in disorder, the parameter initialization function can be used, so that the motherboard parameters will be back to the factory default value.

Parameter Initialization Operation	
Key function	**Menu display**
MODULE + 3 DOWN + 3 DOWN Initialization, XIZI version Password: 654321	PASSWORD ------
△▽ + CLEAR ENTER	XOEC=ON XIZI=Ent RETURN CLEAR
Press △▽ + 1 ON	

The initialization logic curve parameter is the XOEC parameter	RTC-1 PASSWORD=2 INI=3 PART=4
Press △▽ + CLEAR ENTER The initialization logic curve parameter is the XIZI parameter	RTC-1 PASSWORD=2 INI=3 PART=4

1.1.9 Server Password Setting Menu (M-3-2)

This menu can be set with the server login password to prevent others from using the server to change the parameters.

Server Password Setting Operation	
Key function	**Menu display**
MODULE + 3 DOWN + 2 UP	Enter new passwd ------
0 OFF … 9 TEST F Enter new password	Enter new passwd ******
△▽ + CLEAR ENTER Confirm	Confirm passwd ------
0 OFF … 9 TEST F Re-enter new password	Confirm passwd ******
△▽ + CLEAR ENTER Confirm	NEW Password is successful!

2 Debugging Instructions

2.1 Maintenance Mode Running Condition Check

Tip: before activating slow speed, ensure that all the mechanical parts have been debugged, please refer to the specific installation instructions.

2.1.1 Check the control cabinet

Open the door of the control cabinet, check if any connection loosens and any component damaged, well keep the data attached, replace the damaged parts, and fasten all the joints in the control cabinet. When fastening, please pay special attention to the connection of the power cord, power line and brake resistance line.

2.1.2 Check the connection

According to the wiring diagram, check the PVT line, the temporary wiring of the accompanying cable and the temporary wiring of the limit switch, and check whether the grounding wire of all equipment has reliable grounding.

2.1.3 Insulation check

Disconnect the grounding wire and HL, pull out all the plug-ins on SMART, pull out related plug-ins of all the doors, calls, instructions and display signals, put the air switchesat "OFF" position, and use the insulation tester to measure the insulation resistance value ofthe two ends of grounding wire and HL, power line, motor power line, safety loop, control circuit, brake coil, door machine and lighting, to ensure the insulation resistance value is within the prescribed value, and re-connect the ground wire and plug-ins on SMART board.

Circuit	**Allowable insulation resistance**
Power circuit and safety circuit	≥ 0.5MΩ
Control circuit (including door machine) , lighting circuit and signal circuit	≥ 0.5MΩ

Note: in the test of insulation resistance, be sure to remove the plug on the electronic board, otherwise it may damage the electronic board.

2.1.4 Check the input voltage

Cut off the main power air switch and other air switches in control cabinet, check whether the three-phase input voltage is within the specified range (± 10%), and according to the actual input voltage, adjust the connection of the transformer input terminal (if the input voltage is below 370V, then connected to 360V; if the input voltage is 371V~390V, connected to 380V; if the input voltage is 391V~410V, then connected to 400V; if the input voltage is above 411V, then connected to 415V), check whether the voltage (lighting voltage) of C16 and C17 ends on the terminal block is 220V ±10%, and check whether the power indicator on driving component is normal (power indicator is in the square hole below the SMART board).

2.1.5 Check the output voltage of the control transformer

Close the main power air switch, and check whether the output terminal voltage of the transformer is consistent with the drawings (transformer output is allowed to have ±10% error when the input voltage meets the requirements).

2.2 Power-on check

2.2.1 Power-on check the SMART state

Cut off the main power switch, and plug in all the plug-ins.

Check the status of the CHC and DDO signal lights on the SMART electronic board.

Switches S4 and S6 have the following features:

CHC indicator-- up: Cancel the hall call;

-- out: Normal operation

DDO indicator -- up: Cancel door operation;

-- out: Normal operation

Short blocks J1, J2, J8, J10, J11, J12, J15 and J16 have the following functions:

J1 driver debug port (driver board):

J2 encoder power selection (5V, 8V)

J8 J10 J15 J16 communication terminal absorption selection

J11 J12 (driver board) encoder type selection, the two pins on the right of the short circuit represent the use of incremental encoder; the two pins on the left of the short circuit represent the use of cosine encoder

Make sure that the ERO switch on the control cabinet is in an emergency operation position.

Make sure that all hall doors and car doors are completely closed.

Switch on the main power

Observe the indicator light on the SMART electronic board and check whether the input signal is correct or not:

Indicator light	Description
+5V	Light up: power is normal (5V)
RSL	Flash: communication is normal
LVC	Light up: bypass function is opened
DZ	Light up: car is within the door area
DOB	Light up: door inversion device (front door or rear door) is operated
DOL	Light up: limit switch (front door or rear door)
DCL	Light up: limit switch (front door or rear door)
DFC	Light up: car door and safety chain closed up
DW	Light up: hall door closed up
ES	Light up: safety loop disconnected
INS	Light up: elevator in maintenance state

NOR	Light up: elevator in normal state
GRP	Light up: elevator in group control state
DRV	Light up: main board and driver communication normal

Note: if the status of the indicator light is not consistent with the state listed in the table after the power on, please check the relevant circuit and parameters (usually parameters have been set in the elevator factory).

All the installation parameters and I/O parameters of SMART have been set in the factory.Please refer to the relevant parameter table of SMART in detail.

If necessary, confirm the following parameters:

Parameter	Parameter Description	Set Value	Remarks
M-1-3-1-1			
TOP	Highest floor	Set according to the contract	Calculate from 0
BOTTOM	Lowest floor	0	
ERO-TYP	ERO type definition	1	
M-1-3-1-4			
DRIVE	Drive type	0	Set as 1 if there is the advanced door opening function
M-1-3-1-5			
DOOR	Front door machine type	5	DO/DC is 5, DT is 14
REAR	Rear door machine type	0	
M-1-3-1-6			
EN-RLV	Re-leveling setting	0	Set as 1 if there is the re-leveling function
M-1-3-1-10			
NoDW_Chk	DW signal check selection	0	
EN-EVT	EEPROM is allowed to keep error record when power off	1	
EN-CRT	Contract operation enable	3	Set as 0 if the number of floor is 16, set as 1 if it is 32, set as 2 if it is 48 and set as 3 if it is 64

2.3 Parameter set of drive part

2.3.1 Insert the server into the SVT interface, M211 monitors curve status, M212 monitors drive state, and see 5.3.14 curve monitoring menu and 5.3.15 drive monitoring menu for specific menu contents.

M214 records the self learning position information, and see the 5.3.16 self-learning position display.

M23 monitor fault records, and see the 5.3.19 fault list

M221-227 for parameter set;

M33 will do initialization for all parameters, please use this menu carefully;

M24 will carry on the well path position self-learning;

For elevator installed with synchronous motor, it is a must to take the host automatic positioning and self learning of the well. Before the self positioning and the well self learning, set the motor parameter and encoder parameter, enter into the M2-2-1 and M2-2-5 menu to set motor parameter and, input the specific data in accordance with the motor nameplate used at the site.

Field parameters adjustment enters the M2-2-1 (factory parameters take 1.75m/s, 11.7KW host, for example)

Key Value	Display Content	Chinese Explanation	Factory Value	Set Value Range
M2-2-1	Shv diam mm	Pulley diameter	400	10~2000
	Gear ratio	Reduction ratio	1.0	1.0~100.0
	Rope ratio	Rope winding ratio	2	1~6
	Rated RPM	Rated rotation speed	167	10~9999
	Encoder PPR	Encoder pulse number	2048	500~10000
	Rotate Dir	Rotate direction	0	0~1
	Brk Sw Pres? 1/0	Brake switch detection	1	0~1
	overspeed (PU)	Overspeed percentage set, when it is 0, shield the function	110	0~150
	Brk dect dlay ms	Brake switch detection delay	600	0~9999
	Dely brk lftd ms	Delay between the time when brake contactor instruction is issued and the time when the speed is given	500	0~9999
	Delay lft brk ms	Delay from the output contactor instruction to the brake contactor instruction	1000	300~2000
	0 mm/s t lim ms	Delay from the time when speed instruction reaches 0 to the time when brake contactor issues the command	0	0~5000

Speed Regulator Parameter Adjustment Enters M2-2-2

Key Value	Display Content	Chinese Explanation	Factory Value	Set Value Range

M2-2-2	Velocity normal	Rated speed	1748	10~duty speed
	Accelera normal	Normal acceleration set	600	10~1500
	Acc Jerks mm^3	Acceleration J-START	500	20~1500
	Acc Jerke mm^3	Acceleration J-END	500	20~1500
	Decelera normal	Normal deceleration set	600	10~1500
	Dec Jerks mm^3	Deceleration J-START	500	20~1500
	Dec Jerke mm^3	Deceleration J-END	350	20~1500
	Velocity inspect	Inspection speed set	200	10~630
	Decelera rescue	Nearest leveling decelertion	180	10~1500
	Velocity learn	Self learning speed set	100	10~500
	Accelera learn	Self learning acceleration set	600	10~1500
	Velocity relevel	Re-leveling speed set	30	10~100
	Accelera relevel	Re-leveling acceleration set	300	10~1500
	Decelera NTSD	Forced deceleration	1300	10~1500
	Jerk NTSD mm^3	Forced deceleration JERK value set	200	10~1500
	pos delay (ms)	Position delay	100	0~300
Key Value	**Display Content**	**Chinese Explanation**	**Factory Value**	**Set Value Range**
M2-2-2	Run source SVT=0	Manual run enable	1	0~1
	Run enable	Fast run enable	0	0~1
	Door open speed	Advanced door opening speed set	100	0~300
	position gain	Improve the advanced door opening efficiency	20	10~40
	position correct	Elevator position is lost when it exceeds the valve value	200	100~1000
	ARD Speed (mm/s)	ARD running speed set	160	10~200
	Reset Speed %	Reset running percentage set	50	10~100
	ARD Run Direct	ARD running direction set	0	0~1
	Reset At PowerOn	1: power-on reset running 0: non-reset running	0	0~1
	Inspect 0stop En	Maintenance 0 speed stop enable	1	0~1 1: 0speed stop 0: emergency stop
	ETSC percent %	ETSC ction percentage set	94	0~200
	ETSC Enable(0/1)	ETSC action enable	0	0~1

Note: in asynchronous machine debugging, Accelera normal, Acc Jerks mm^3, Acc Jerke mm^3, Decelera normal ,Dec Jerks mm^3 are generally set as 150-250; Dec Jerke mm^3 50-150.

M2-2-3 set leveling parameters

Key Value	Display Content	Chinese Explanation	Factory Value	Set Value Range
M2-2-3	Up level	Upgoing leveling parameter	000	0~500
	Down level	Downgoing leveling parameter	000	0~500

M2-2-4set starting torque parameters

Key Value	Display Content	Chinese Explanation	Factory Value	Set Value Range
M2-2-4	Inertia kg/m2	System inertia	30.0	0.1~999.9 Note: the asynchronous machine is set below the range of 10
	Start Kp	Proportional coefficient of pre torque calculation	3500	0~9999 Note: the asynchronous machine is set as 0
	Start Ki	Pre torque calculation integral coefficient	350	0~9999 Note: the asynchronous machine is set as 0
	APR time(ms)	Pre torque calculation time	200	10~900
	Brk dect dlay ms	Read-only	600	300~2000
	0mm/s t lim (ms)	Read-only	0	0~5000
	Brake settle(ms)	Read-only	1300	300~5000

The motor parameters are in M2-2-5, and input the specific value according to the field host nameplate (take 11.7KW host for example):

Key Value	Display Content	Chinese Explanation	Factory Value	Set Value Range
M2-2-5	Control methord	Control mode of main engine	3	3 for the synchronous machine, 1 for the asynchronous machine
	Encoder PPR	Read-only	2048	
	Encoder Sort	Encoder type	1	0: square wave increment 1: sine cosine
	Rated Power(Kw)	Rated power of	11.7	0.1~999.9

		main engine		
	Number of poles	Number of poles of main engine	24	1~99
	Rated RPM	Rated rotation speed of main engine	167	10~9999
	Rated frq (Hz)	Rated frequency of main engine	33.40	0.01~99.99
	Rated voltage(V)	Rated voltage of main engine	340	10~999
	Duty load (kg)	Rated load of main engine	1000	10~9999
	Rated I (A)	Rated current of main engine	26.0	0.1~999.9
	Rated Trq (Nm)	Rated torque of main engine	669	1~9999 Note:in the use of asynchronous machine, if the host nameplate does not indicate the parameters, it is set as about 80.
	Shv diam (mm)	Read-only	400	10~10000
	Gear ratio	Read-only	1.0	1~100
	Rope ratio	Read-only	2	1~6
	mutual resist	Main engine resistance	0.41	0~99.99
	induct d0(mH)	Main engine static d axis inductance	13.00	0~99.99
	induct q0(mH)	Main engine static q axis inductance	13.00	0~99.99
	induct d(mH)	Main engine operating d axis inductance	8.00	0~99.99
	induct q(mH)	Main engine operating q axis inductance	8.00	0~99.99
	Rotor Time (s)	Rotor time constant of induction motor	0.28	0~10
	No load current	No-load current of induction motor	11.6	0.1~999.9

Main engine inductance parameter reference value:

Motor specification	**Ld (mH)**	**Lq (mH)**	**Lq0 (mH)**	**Ld0 (mH)**

340V,29A,239RPM,16P	5	5	10	10
340V,30A,127.5RPM,16P	5	5	10	10
340V,26A,167 RPM,12P	8	8	13	13
340V,26A,209RPM,12P	8	8	13	13
340V,21A,167 RPM,16P	8	8	13	13

Note: if the main engine nameplate does not indicate the inductance value, please refer to this table. In general, when rated current increases, the inductance value decreases.

Recommended calculation formula of main engine inductance parameter:
d = 80/main engine rated power
q=d
r = 4.2/main engine rated power
d0 = 1.5 * d
q0 = 1.5 * q
Note: 1.main engine rated power unit: KW;
2.main engine rated power parameter name is Rated Power(Kw)。
For example, 10kW main engine, d and q is set as 8, d0 and q0 is set as 12.

Inertia calculation formula:
Inertia = ((rated load *3 +700)/ rope winding ratio/ rope winding ratio+125) *traction wheel diameter * traction wheel diameter /4
Note: 1. load unit is kg, and diameter unit is meter;
2. inertia parameter name is Inertia kg/m2, rated load parameter name is Duty load (kg), rope winding ratio parameter name is Rope ratio, and traction wheel diameter parameter is named as Shv diam mm;
For example: load is 1000kg, rope winding ratio is 2, traction wheel diameter is 400mm, and the inertia is 42.

Corresponding to different drive bases, the parameter Drive size setting values are as follows:

Driver Power	Motherboard Driver Parameter Value	Base Model	Remarks
7.5kw	5	SMART100-LR-4007-H3	
11kw	6	SMART100-LR-4011-H3	
15kw	7	SMART100-LR-4015-H3	
18.5kw	8	SMART100-LR-4018-H3	
22kw	9	SMART100-LR-4022-H3	
30kw	10	SMART100-LR-4030-H3	
37kw	11	SMART100-LR-4037-H3	

Note: Drive size parameter value is factory default, matching with the driver and can not be changed.

Drive part debugging

2.4.1 Running status settings

M222 Run enable	M222 Run source	Allowed state
0	0	Parameter initialization, manual operation
0	1	Self learning and maintenance operation of well
1	1	Maintenance operation, fast operation and, reset operation

2.4.2 Debugging when using synchronous motor

For the synchronous motor, before the drive part debugging, press the maintenance up or down button for the main engine static automatic positioning. The overall debugging process of the drive part can be conducted according to the steps of setting the encoder parameters, setting the driver type, and setting the parameters of the main engine.

2.4.2.1 Static automatic positioning of synchronous main engine encoder

1. Set the RUN SOURCE SVTin M2-2-2 menu as1.

2. Enter the correct parameters: M2-2-1 encoder parameters, and set the Control mothed in M2-2-5 menu as 3, to determine the main engine parameters in M2-2-5.

3. Set the Encoder Dir as 0, and in the use of V1 line, UVW phase is corresponding to each other, and in the use of V2 line, two of the UVW phase can be mutually changed. Monitor the server 2-1-2Output current A, hold the computer room maintenance ascending or descending button, the drive takes automatic positioning, and positioning process lasts 3-5 seconds (hold the maintenance ascending or descending button throughout the positioning process), the main engine sends a humming sound, when the main engine power is less than 6kW, Output current value should be about 30% rated current; when the main engine power is larger than 6kW, Output current value should be about 10% rated current. After the end of the positioning process, the motor starts running, if the motor runs smoothly, the speed feedback is correct, then no-load Output current should be below1A. If the motor jitters and has Overcurrent fault, you will need to change any two UVW phase, and if it runs normally after re positioning, observe the running direction, if it is opposite to the actual direction, then change the rotation direction2-2-1Rotate dir; when running upward, check 2-1-1Now Position m and the system is normal when the monitoring value increases.

6.4.2.2 Converter jumper and encoder connection check

Synchronous motor uses the German HEIDENHAIN encoder (sine cosine rotary encoder, with reference mark and two sine cosine absolute position sensors). For different encoder, encoder power jumper J2 setting of SMART board is not the same, 5V or 8V can be chosen according to the encoder working voltage. The connecting cable of SMART board jumper and the encoder is ready at the factory, on finding the server display encoder fault or the motor can't operate normally, please check these two items while ensuring the parameter setting is correct.

Encoder cable is connected as shown in the following table:

Encoder interface	Line color	Interface	External PIN
1a	Pink (C-)	Sin- Abs	10
6a	Purple (D-)	Cos- Abs	13
2a	Yellow/black(A-)	Sin- Inc	6
5a	Red/black (B-)	Cos- Inc	1
7b	Grey (C+)	Sin+ Abs	11
2b	Yellow(D+)	Cos+ Abs	12
6b	Green/black(A+)	Sin+ Inc	5
3b	Blue/black(B+)	Cos+ Inc	8
1b	Brown/green(+5V)	+5V	9
5b	White/green (0V)	0V	7
4a	Black (R-)	Zero-	4
4b	Red (R+)	Zero+	3

2.4 Elevator running direction check

Enter the maintenance operation mode

In jog running, it is necessary to confirm the operation directions of the elevator are the same, if not consistent, it is necessary to adjust.

In jog running of the elevator, make sure the actual running direction of the elevator is consistent with the direction of maintenance instruction.

If jog running upward, but the elevator actually runs downward (or the opposite), it indicates the running direction of the elevator is opposite to the maintenance instruction, then enter the smart board menu M-2-2-1, reversely take the Rotate dir parameters.

After saving the parameters, make sure the direction is consistent.

Jog running of the elevator, enter the smart board menu M-2-1-1 for monitoring Now position m.

If jog running upward, and the display value is decreasing (or the opposite), it indicates that the pulse counting direction is opposite, enter the smart board menu M-2-2-1, and reversely take the Pulse Cnt Dir parameter.

After saving the parameters, make sure the elevator running direction is consistent with the pulse calculation direction.

2.5 Jog running mode

2.6.1 Electrical recall operation (ERO)

Make sure the control cabinet ERO switch is in the service position, and the car top maintenance switch (TCI) is in the normal position.

Enter (M-1-1-2) into the server to observe the tci, uib, dib, ero state, at this time the server should show "tci, uib, dib, ERO".

Jog the ascending button on ERO box, the uib in the server will turn into the capital UIB.

Jog the descending button on ERO box, the dib in the server will turn into the capital DIB.

Keep pressing the ascending button and confirm the elevator runs upward.

Keep pressing the descending button and confirm the elevator runs downward.

2.6.2 Car top maintenance operation (TCI)

Put the car top maintenance switch (TCI) to the maintenance position, and the emergency electric switch (ERO) of the control cabinet in the machine room to the normal position. (Note: when operating the maintenance switch on car top, it is a must to follow the "car top operating procedures", otherwise the logic control part will protect and the elevator will not run, then the server will display "TCI-LOCK" flashing information, also the INS indicator on the electronic board of logic control part flashes.)

At this time the tci in the server turns into the capital TCI.

Press the ascending button U and C simultaneously to make sure the elevator is running upward.

Press the descending button D and C simultaneously to make sure the elevator is running downward.

Carefully let the elevator run in the well in maintenance state, and ensure that no prominent obstacles in the well stop the operation of the elevator. If there is any, then take the appropriate measures.

Check on the car top whether TES (car top emergency stop switch), EEC (safety window switch), SOS (safety clamp switch) and upper and lower limit switch function is effective.

2.6 Position reference system adjustment

2.7.1 Limit switch adjustment

According to the table below, adjust the limit switch distance (the allowable error of these distance shall not be more than 20 mm). The positive and negative signs in front of the numerical values are determined: the leveling position when the elevator is at thetop and ground floors is taken as the benchmark, mark at the guide rail as 00 mm.

For top floor, positive sign means above the mark, and negative sign means below the mark.

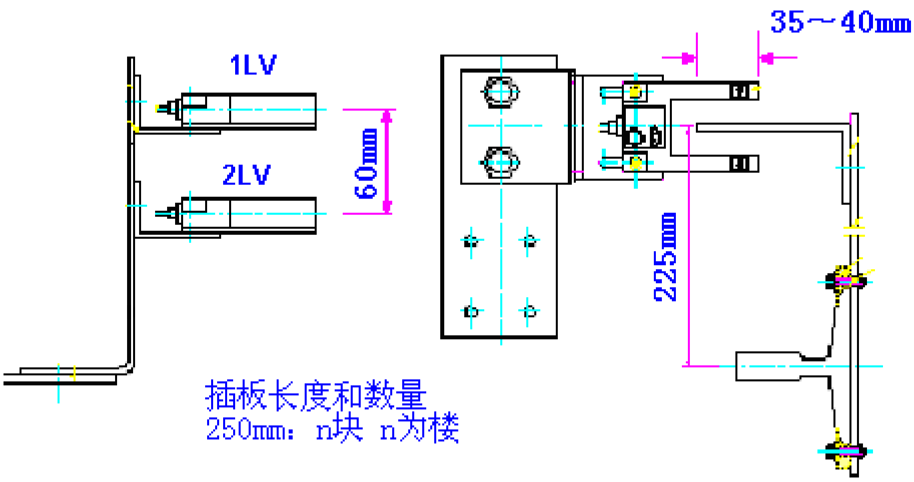
For ground floor, positive sign means below the mark, and negative sign means above the mark.

Speed(m/s)	1LS,2LS(mm)	5LS,6LS (mm)	7LS,8LS(mm)
0.5	-350	+50	150±50
0.75	-570	+50	150±50
1.0	-840	+50	150±50
1.50	-1610	+50	150±50
1.75	-1800	+50	150±50
2.00	-2190	+50	150±50
2.50	-3320	+50	150±50

Note: the distance referred to here is the distance when the limit switch contact is opened, rather than the distance when the limit switch roller presses on the linkage.

2.7.2 Adjust the car top photoelectric switch and well light partition board

Mounting method and mounting dimension of the car top photoelectric switch and partition board are shown in the figure below.



Put into the car the balance load (about 45% load)

Adjust the photoelectric switch position on the two leveling (1LV: upper leveling photoelectric switch; 2LV: lower leveling photoelectric switch), so they are about 60mm away and vertical, and, ensure the installation sequence of photoelectric switch is1LV and 2LV from top to bottom.

Run the elevator to the leveling position of each floor.

Adjust the partition board of each floor to make the central line and the center line of the two photoelectric consistent (i.e. the center line is 30mm from 1LV and 2LV respectively). This operation will affect the leveling accuracy of the elevator.

2.7 Prepare for the first time normal operation

2.8.1 Safety and lock loop check

Make sure each safety switch of the safety loop is effective and in the maintenance operation conditions, turning on every safety switch (OS, 8LS, 7LS, SOS, TES, PES, GS, DS, GSS) is able to make the elevator stop (note: please confirm the lock of landing door on each floor is effective).

2.8.2 Confirm the well signal

Use TCI or ERO to operate the elevator all the way, check the door area 2LV and 1LV, and forced deceleration 1LS, 2LS and other well signals: use the server to enter into the M1-1-2 to check the input signal of SMART board;

2.8 Well location self learning

1) Before the well location self learning, carry out full operation through ERO, and use the server to observe and confirm the well photoelectric and forced deceleration switch signal is normal.

2) When the elevator is in the leveling position, DZ, 1LV and 2LV should be capitalized; in downward movement, the photoelectric first takes action, so DZ and 2LV turns into lower case, and it is contrary when running upward.

3) When the elevator is near the bottom floor, 1LS action, so 1LS should be capitalized; when the elevator is at the top position, 2LS action, so 2LS should be capitalized; when the elevator is on the middle floor, 2LS and 1LS should be lowercase.

4) Confirm the value of TOP in M-1-3-1-1 SMART, and set the RUN ENABLE parameter in M-2-2-2 as 0, RUN SOURCE SVT as 1.

5) Switch ERO and TCI to the normal position, and use the server to operate the drive part M-2-4, and press Shift+Entir to start the well self learning.

6) The elevator will run to the leveling position on the bottom floor at low speed, and then run upward for well self learning at self learning speed, when it reaches the top level, self learning is completed.

7) After the success of the self learning, change the RUN ENABLE parameters in M-2-2-2 to 1, and

the well location information is stored in the M-2-1-4 menu, please check whether it is correct.

8) Switch ERO to the maintenance position, press the descending button to run downward for some time and make the car into non leveling position, and then put ERO to the normal position, at this point, the elevator should take reset running, monitor M-1-1-1 with server, it should display the COR status, until reset to the leveling position.

2.9 Normal operation

2.10.1 According to the wiring diagram, check whether the remote station RS5 address has any error and the pin connection is correct.

2.10.2 Connect the server to the SMART board SVT server interface, and according to the parameter table and the I/O port input and output table, confirm all theparameters and the input and output addresses within EEPROM is correct.

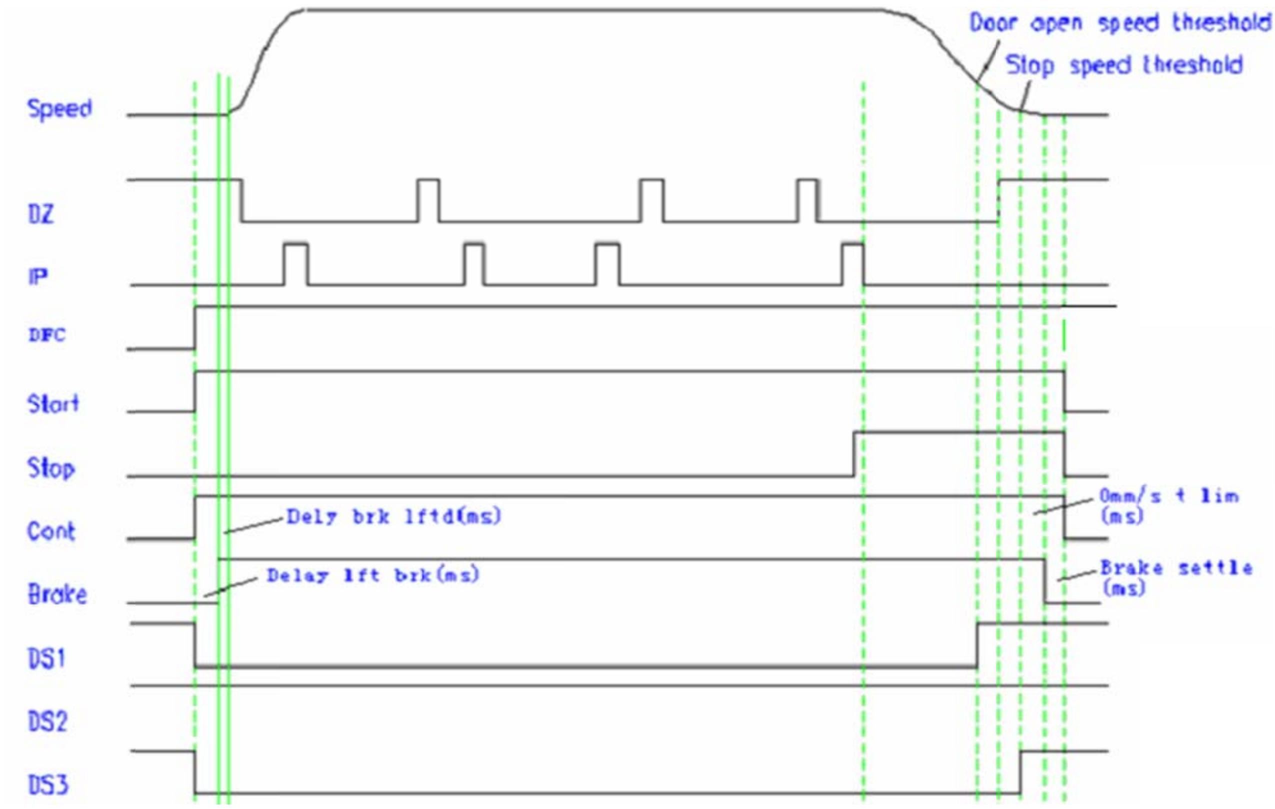
2.10.3 Use server to enter call signal

2.10.4 Press the server M-1-1-1 in turn, and then enter the call signal, and press the number of floor required. (0 for 1st floor, 1 for 2nd floor, and so on)

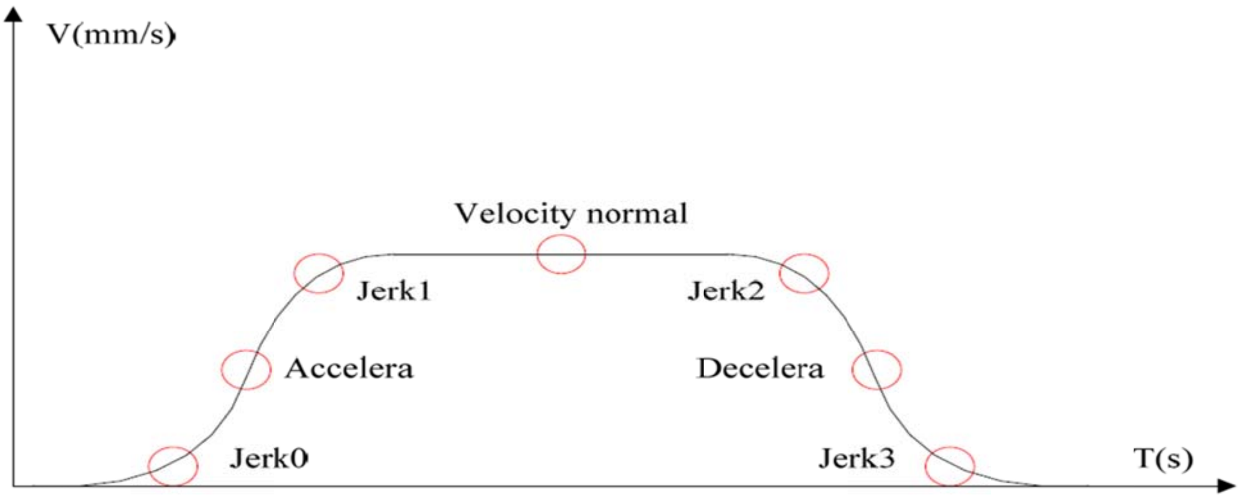
2.10.5 Press the blue button (shift), and then press the "ENTER" button, the elevator will run to the calling floor.

2.10.6 Check to make sure all functions related to RS5 and RS32 are normal.

2.10.7 Use server to monitor the Car speed in M-2-1-1, run the elevator all the way, and check whether the elevator is running at the contract speed.



Normal operation sequence diagram of elevator



Normal operation curve of elevator

2.10 Adjustment of leveling position in normal operation

2.11.1 Prior to the electrical leveling adjustment, ensure that the mechanical door panel has been adjusted.

2.11.2 Run downward and record the leveling error of each floor, and according to the average value of error, correct the DOWN LEVEL value in M-2-2-3. (Reduce the value if it passes, and vice versa).

2.11.3 Run upward and record the leveling error of each floor, and according to the average value of error, correct the UP LEVEL value in M-2-2-3. (Reduce the value if it passes, and vice versa).

2.11.4 Adjustment of leveling position is completed.

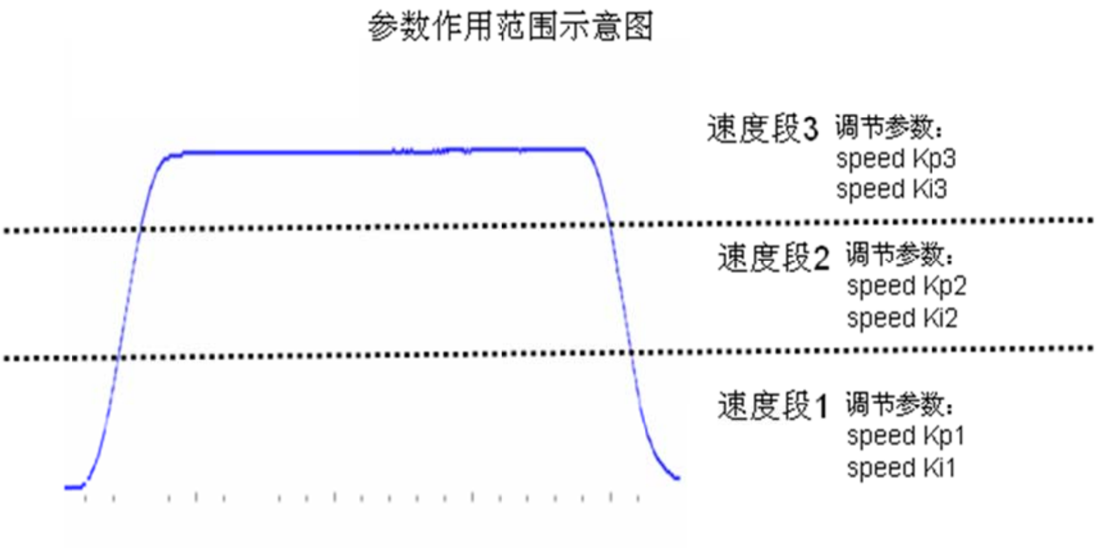
2.11 Start the adjustment of comfort

Put the elevator in CHC and DDO state, fast run the elevator and observe whether there is any jitter or slide when traction wheel starts up, if any jitter is observed, then it indicates the parameter matching performance is not good, you can enter the M-2-2-4 to adjust the system Inertia kg/m2, increase or decrease 5 each time, generally, the elevator will have high-frequency jitter when it stops if the system inertia is too big, while if it is too small, the elevator will be underpowered at startup and have sliding. If the adjustment of system inertia produces no obvious change, then adjust the M-2-2-4 Start Kp and Start Ki at the same time, in general, if the rigidity of the system is strong (no car top spring), set as Kp3000~7000, Ki150~700; if there is the car top spring generally set KP1000~3500, KI100~350; KP and KI has the best correlation values in this range. If there is still the sliding after Kp and Ki is adjusted, you can adjust M2-2-4 APR time when there is the main engine brake open delay phenomenon, APR time is the duration of the output torque compensation, generally APR time has the best correlation value between 100~400. Adjust the above parameters, and observe the elevator condition, until it reaches the best condition.

2.12 Adjustment of running comfort (low-frequency jitter at high speed)

Put the elevator in CHC and DDO state, fast run the elevator to observe whether the traction machine has any jitter from low speed to high speed, if any jitter is observed, then enable the speed regulation. Enter the M-2-2-4 to adjust speed low sect and speed mid sect value, the speed will be divided into three speed segments, when the elevator speed is less than the speed low sect percentage rated speed, it is low speed segment (speed segment1), when it is more than speed low sect and less than speed mid sect percentage rated speed, it is middle speed segment (speed segment2) and when it is more than speed mid sect percentage rated speed, it is high speed segment (speed segment 3).

For example, to adjust the high-speed segment (speed segment 3) jitter problem, you need to make simultaneous adjustment of speed Kp3 and speed Ki3. Speed Kp3 and speed Ki3 has the best correlation value in adjustable range (default value 1000, adjustable range 10~10000), simultaneous increase or decrease in adjustment, and each change value is 200; if the main enginehas obvious abnormal jitter in the adjustment process, appropriately reduce the parameter until the operation achieves the best condition.



2.13 Board server debugging instructions

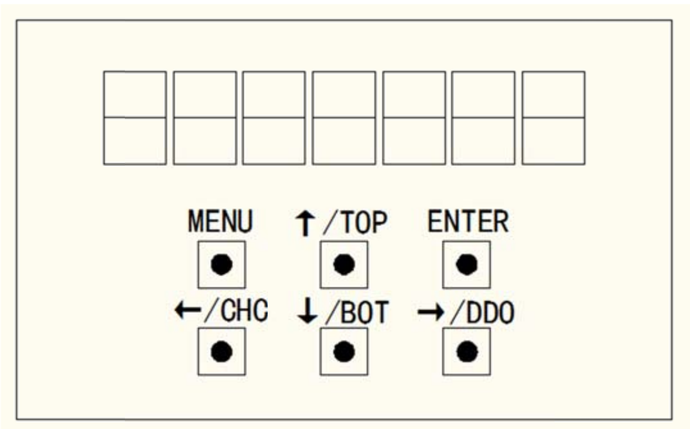
Board server consists of seven bit digital tubes and six bit keys. It mainly has the function of realizing the real-time state display of the controller, the partial parameter setting, the running times display and so on.

Board server has exposed structure, please pay attention to the following in use:

Caution
◆ In order to avoid the electric shock accident, or damage to the control panel device due to human body electrostatic, please wear insulated gloves for operation; ◆ It can't be operated by metal or sharp tools, to avoid short circuit fault or damage to components on the board.

Basic function description

Small keyboard appearance



Functions of keys on small keyboard

Six keys are MENU, ↑/TOP, ENTER, ←/CHC, ↓/BOT, →/DDO

Key	Name	Function
MENU	Menu key	In any state, press the MENU button to display the current function menu, and you can switch function menu with ↑/TOP and ↓/BOT key
↑/TOP	Increasing key	In the functional group menu, you can page down the menu through ↑/TOP key. In the parameter settings menu, you can set the value of the relevant parameters by the key. In addition, in specific function menu (under Sn shortcut menu), you can achieve TOP key functions (top floor command).
↓/BOT	Declining key	In the functional group menu, you can page up the menu through ↓/BOT key. In the parameter settings menu, you can set the value of the relevant parameters by the key. In addition, in specific function menu (under Sn shortcut menu), you can achieve BOT key functions (bottom floor command).
←/CHC	Left shift key	Left shift key in parameter setting menu; after setting up the current parameters, enter the next setup by←/CHC key. In addition, in specific function menu (under Sn shortcut menu), you can achieve CHC key functions (landing floor call enable / disable).
→/DDO	Right shift key	Right shift key in parameter setting menu; after setting up the current parameters, enter the next setup by→/DDO key. In addition, in specific function menu (under Sn shortcut menu), you can achieve DDO key functions (door opening and closing operation enable / disable).
ENTER	Confirm key	In the functional group menu, enter the functional group data menu through ENTER key. In the parameter settings menu, set the parameters and then press ENTER key to save the value of the parameters, and display the next parameter settings interface.

Composite key function

In the specific interface (En fault menu), press the ←/CHC key and →/DDO **key at the same time, to clear the current and historical fault of drive part.**

Menu description

First level menu	**Second level menu**	**Third level menu**

Sn shortcut menu	“1234567” operation times	None
Pn monitoring menu	P0	P0-00~P0-09
Fn setting menu	F0~F3	FX-00~FX-XX
Ln logical parameter	L0	
En fault menu	E0~E20	
An password	A0~A1	
Cn self learning	Lrn	
Current fault		

Sn shortcut menu

This menu shows the total number of operation, and the button function is its second function (see "description of small keyboard button function ").

Pn system status

P0 operating curve

Serial No.	Content	Description
P0-00	NOW FLOOR	Current floor (the right button value can adjust the call floor (range 0~top))
P0-01	NOW POSITION m	Current location m (two decimal places)
P0-02	CAR SPEED mm/s	Actual speed of elevator mm/s
P0-03	DEDICATED mm/s	Speed given mm/s
P0-04	DUTY SPEED mm/s	Rated speed of elevator mm/s
P0-05	ROTATE SPD RPM	Rotate speed of motor RPM
P0-06	Output current A	Output current A
P0-07	Output voltage V	Output voltage V
P0-08	Heat sink temp1	Driver temperature
P0-09	Dc link V	Bus voltage

Fn setting menu

F0-3run common parameter setting

Parameter No.	Parameter	Default	Range	Description
F0-0	Rated Power(Kw)	11.0	0.1~999.9	Main engine rated power

F0-1	Number of poles	24	1~99	Main engine pole number
F0-2	Rated RPM	167	10~9999	Rated rotate speed
F0-3	Rated frq (Hz)	33.40	0.01~99.99	Main engine rated frequency
F0-4	Rated voltage(V)	340	10~999	Main engine rated voltage
F0-5	Duty load (kg)	1000	10~9999	Main engine duty load
F0-6	Rated I (A)	1000	0.1~999.9	Main engine rated load
F0-7	Rated Trq (Nm)	412	1~9999	Main engine rated torque
F0-8	Shvdiam mm	400	10~2000	Drive sheave diameter
F0-9	Gear ratio	1.0	1.0~100.0	Reduction gear ratio
F0-10	Rope ratio	2	1~6	Rope winding ratio
F1-0	Inertia kg/m2	30.0	0.1~999.9	System inertia
F1-1	Rotate Dir	0	0~1	Rotate direction
F1-2	Encoder PPR	2048	500~10000	Encoder pulse number
F1-3	Start Kp	3000	0~9999	Pre torque calculation proportional coefficient
F1-4	Start Ki	300	0~9999	Pre torque calculation integral coefficient
F1-5	APR time(ms)	200	10~900	Pre torque calculation time
F1-6	Pretorque delay	400	0~1000	Pre torque output delay
F1-7	Brkdectdlayms	600	0~9999	Brk_hold_delay Delay from the time after brake down instruction is issued to the detection brake feedback
F1-8	Delybrklftdms	500	0~9999	Delay_brk_lftd delay from the time the brake contactor closing instruction is issued to the time the speed is given
F1-9	Delay lftbrkms	1000	300~2000	Delay_lft_brk delay from the output contactor closing instruction to the brake contactor closing instruction
F1-10	0 mm/s t limms	0	0~5000	Delay from the time when speed instruction reaches 0 to the time when the brake contactor releases instructions

F1-11	Brake settle ms	1300	300~5000	Brk_settle_delay delay from the brake contactor releases instruction to the time when ptr is set as 0
F2-0	Velocity normal	1748	10~duty speed	Rated speed
F2-1	Accelera normal	600	10~1500	Normal acceleration setting
F2-2	Acc Jerks mm^3	500	20~1500	Acceleration J-START
F2-3	AccJerke mm^3	500	20~1500	Acceleration J-END
F2-4	Decelera normal	600	10~1500	Normal deceleration setting
F2-5	Dec Jerks mm^3	500	20~1500	Deceleration J-START
F2-6	Dec Jerke mm^3	350	20~1500	Deceleration J-END
F2-7	Velocity inspect	200	10~630	Maintenance speed setting
F2-8	Decelera rescue	200	10~1500	Nearest level deceleration
F2-9	Velocity learn	100	10~500	Self learning speed setting
F2-10	Velocity relevel	30	10~100	Re-leveling speed setting
F2-11	Door open speed	100	0~300	Advanced door opening speed setting
F2-12	position gain	15	10~40	Improve the advanced door opening efficiency
F2-13	ARD Speed (mm/s)	160	10~200	ARD running speed setting
F2-14	ARD Run Direct	0	0~1	ARDrunning direction setting
F2-15	Distance comp	0	0~99	Level compensation
F2-16	Up level (mm)	70	0~500	Upper level adjustment
F2-17	Down level (mm)	70	0~500	Lower level adjustment
F3-0	Clear Error	0	0~1	Clear drive error
F3-1	Run enable	0	0~1	Fast run enable
F3-2	Pulse Cnt Dir	1	0~1	Distinguish left and right
F3-3	Reset At PowerOn	1	0~1	1:power on reset running 0: non-reset running
F3-4	Inspect 0stop En	1	0~1	Maintenance zero-speed stop enable,1: zero speed stop 0: emergency stop
F3-5	BrkSw Pres? 1/0	1	0~1	Brake switch inspection

L0 logic common parameter setting

Enter L0 data menu through the MEN, UP, DOWN and ENTER key, select the parameter code to be set as required, the parameter code table is shown below:

Parameter code	Parameter implication	Set range
L0-00	TOP	1~63
L0-01	LOBBY	0~63
L0-02	BOTTOM	0~63
L0-03	PKS-P	0~2
L0-04	GRP-NO	0~255
L0-05	GROUP	0~8
L0-06	DRIVE	0~8
L0-07	EN-RLV	0~8
L0-08	EFO-P	0~1
L0-09	TPOS 1	1~12
L0-10	TPOS 2	0~1
L0-11	TDELAY	0~255

En error monitoring

Show "Err-xx" when there is an error

Show “no-Err” when completing the page turning or when there is no error.

Error code table is shown below:

Serial No.	Error code	Attribute	Error reset mode	Error type
Err-01	IGBT Fault	Drive	Press MaxErrorB to set the time reset	B
Err-02	Over current	Drive	Press MaxErrorB to set the time reset	B
Err-03	Retain			
Err-04	Overtemp	Drive	Automatic reset	C
Err-05	Retain			
Err-06	Drive overload	Drive	Manual reset	B
Err-07	DC link OVT	Drive	Automatic reset	C
Err-08	DC link UVT	Drive	Automatic reset	C
Err-09	Overspeed	Curve	Press MaxErrorB to set the time reset	B

Serial No.	Error code	Attribute	Error reset mode	Error type
Err-10	PVT lost	Drive	Press MaxErrorB to set the time reset	B
Err-12	DRV Comm timeout	Curve	Automatic reset	C
Err-13	Task orun	Drive	Press MaxErrorB to set the time reset	B
Err-14	Tune Moving	Drive	Press MaxErrorB to set the time reset	B
Err-15	Track error	Drive	Press MaxErrorB to set the time reset	B
Err-16	1LV NORMAL CLOSE	Curve	Manual reset	A
Err-17	2LV NORMAL CLOSE	Curve	Manual reset	A
Err-18	Floor number err	Curve	Manual reset	A
Err-19	PARA. ABNORMAL	Drive	Manual reset	A
Err-20	Sfty chain state	Curve	Automatic reset	C
Err-21	Retain			
Err-22	Retain			
Err-23	Retain			
Err-24	Retain			
Err-25	E2 write err	Drive	Press MaxErrorB to set the time reset	B
Err-26	Retain			
Err-27	Retain			
Err-28	Brake dropped	Curve	Press MaxErrorB to set the time reset	B
Err-29	AC Line imbal	Drive	Press MaxErrorB to set the time reset	B
Err-30	AC Line UVT	Drive	Press MaxErrorB to set the time reset	B
Err-31	Retain			
Err-32	Thyristor module	Drive	Press Max DBD to set the time reset	B
Err-33	DBD PICKUP	Curve	Press Max DBD to set the time reset	B1
Err-34	DDP Fault	Curve	Manual reset	A
Err-35	Brake dropped 2	Curve	Press MaxErrorB to set the time reset	B

Err-36	Brk check Err	Curve	Manual reset	A
Err-37	Retain			
Err-38	Retain			
Err-39	Bus overcurrent	Curve	Manual reset	A
Serial No.	**Error code**	**Attribute**	**Error reset mode**	**Error type**
Err-40	Retain			
Err-41	Retain			
Err-42	InvIuOffst	Drive	Press MaxErrorB to set the time reset	B
Err-43	InvIvOffst	Drive	Press MaxErrorB to set the time reset	B
Err-44	InvIwOffst	Drive	Press MaxErrorB to set the time reset	B
Err-45	Retain			
Err-46	Retain			
Err-47	Base AD Offst	Drive	Press MaxErrorB to set the time reset	B
Err-48	ETSC relay fault	Curve	Press MaxErrorB to set the time reset	B
Err-49	Retain			
Err-50	ctrler comm. Err	Drive	Automatic reset	C
Err-51	Position Lost	Curve	Automatic reset	C
Err-52	Retain			
…	Retain			
Err-60	Retain			
Err-61	Vcode abnormal 1	Curve	Automatic reset	C
Err-62	Vcode abnormal 2	Curve	Automatic reset	C
Err-63	Power Lost	Drive	Automatic reset	C
Err-64	Retain			
…	Retain			
Err-69	Retain			
Err-70	parm. dectBrk	Curve	Automatic reset	C
Err-71	parm. 0spd	Curve	Automatic reset	C

Err-73	parm. encode	Curve	Automatic reset	C
Err-74	parm. RPM	Curve	Automatic reset	C
Err-75	parm. DIAM	Curve	Automatic reset	C
Err-76	parm. Gear	Curve	Automatic reset	C
Err-77	parm. Rope	Curve	Automatic reset	C
Err-80	Custom code Err	Curve	Automatic reset	C

An password authentication and settings

A0: Password authentication

A1: Password settings

Note: 1. Initial password is 1234, and after password authentication you can enter into the Fn menu to change the parameters;

2. The effective operation time after password authentication is 30min, it is required to re verify the password if more than 30min

Cn self learning interface

Cn menu, enter the second-level menu Lrn, change numerical value to 1, and press Enter to start self learning.

2.14 Instructions on integrated ARD function debugging

In the process of using the elevator, if the power supply of the system suddenly cut off, it may cause the passengers to be locked in the car. In view of this situation, the integrated system has designed a power outage emergency operation plan. Both system main circuit and working power supply use UPS power supply for power cut emergency operation.

After the system enters into the ARD mode, it runs at ARD Speed (mm/s) in M-2-2-2, with the elevator light direction, if the initial operation direction has an error, you can change the parameter ARD Run Direct to set the direction (factory value is 0). When a leveling signal is detected, keep the door in an open state and the elevator is not running.

ARD operating is divided into three stages, see the table below:

Server display state	English spelling	Chinese explanation	Stage
EPC	Emergency Power Control	Emergency Power Control	After receiving the power off signal of the power grid, wait for emergency backup power rescue operation.
EPR	Emergency Power Rescue	Emergency Power Rescue	Perform emergency backup power rescue operation.
EPW	Emergency Power Wait	Emergency Power Wait	Complete ARD running, and arrive at the level with the door opened in place, and

			enter the emergency power to wait.

Relevant I/0 parameter setting:

Motherboard	I/O	Default value	Set value
SMART	17(generator signal)	Logical bit=0 address=00 bit=0	Logical bit=0 address=01 bit=0
	1676(virtual floor signal)	Logical bit=0 address=00 bit=0	Logical bit=0 address=60 bit=1
	1677(rear door signal)	Logical bit=0 address=00 bit=0	Logical bit=0 address=60 bit=2

Relevant parameter settings:

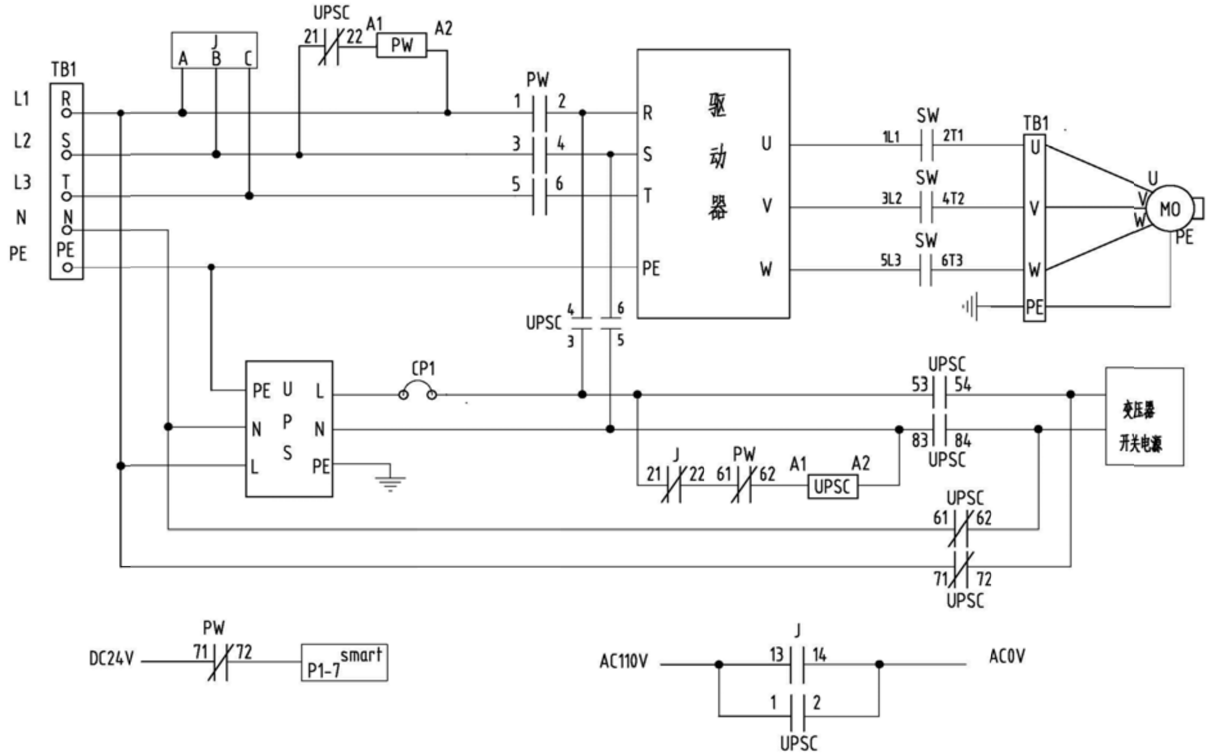
Menu	Parameter name	Factory value	Function and range
M223	ARDRun Mode	1	Function: determine according to the emergency rescue mode; Parameter setting range: 0 - 2; Integrated HSD mode=0; integrated ARD mode=1; integrated HSD+ARD mode=2。
M222	ARD-Speed (mm/s2)	160	Function: emergency standby power rescue operation speed, and at the same time have function on the leveling accuracy in rescue operation. Parameter setting range: 1-500; and according to the leveling accuracy requirements, set range of 80-160 is recommended;
M222	ARD-RUN DIRECT	0	Function: determined according to the left or right motor installation mode; Parameter setting range: 0-1; Motor left installation mode =0; motor right installation mode =1。
M1318	EPO-DC	0	Function: after ARD rescue is completed, open the door to keep waiting time Parameter setting range: 0-255 (S) 。
M1318	EPO-P	64	Function: stop location in EPO mode This parameter must be set to 64 when the integrated ARD function is enabled.
M131-10	EN-EVT	1	Function: whether the motherboard saves logic events after power down 0: not save; 1: save This parameter must be set to 1 when the integrated ARD function is enabled.

ARD function test process:

When all the connection and parameter setting is completed, you can test the ARD function according to the following steps:

1. Ensure that the elevator system is in normal operation mode;
2. Cut off the main power of control cabinet, but the air switch within the control cabinet can't be cut off;
3. Server M1-1-1 monitoring system is in EPC state;
4. When the EPC mode changes to EPR mode, the elevator begins rescue operation;
5. In the ARD mode, the elevator will be running in the light-load direction. For example, when the elevator is in a non-load state, the car is running upward; on the contrary, the car runs downward when the elevator is fully loaded. When the car runs to the door area, the elevator door opens to release the passengers.
6. When the car stops in the door area, the elevator door opens, the server M1-1-1 monitoring system is in the EPW state. You can change the parameter EPO-DC to change the opening wait time (when EPO-DC=0, the door is always open);
7. When the main power of control cabinet recovers, the elevator will be converted to normal operation mode.

Typical circuit diagram design:



2.15 Instructions on reset rescue function debugging

When the elevator has position loss error, it will automatically enter the reset rescue operation mode, and run to the nearest floor at the speed of 300mm/s, and open the door to release the passengers, and then run to the terminal to reset. At the same time, in this process, the voice module inside COP provides voice appease and prompt.

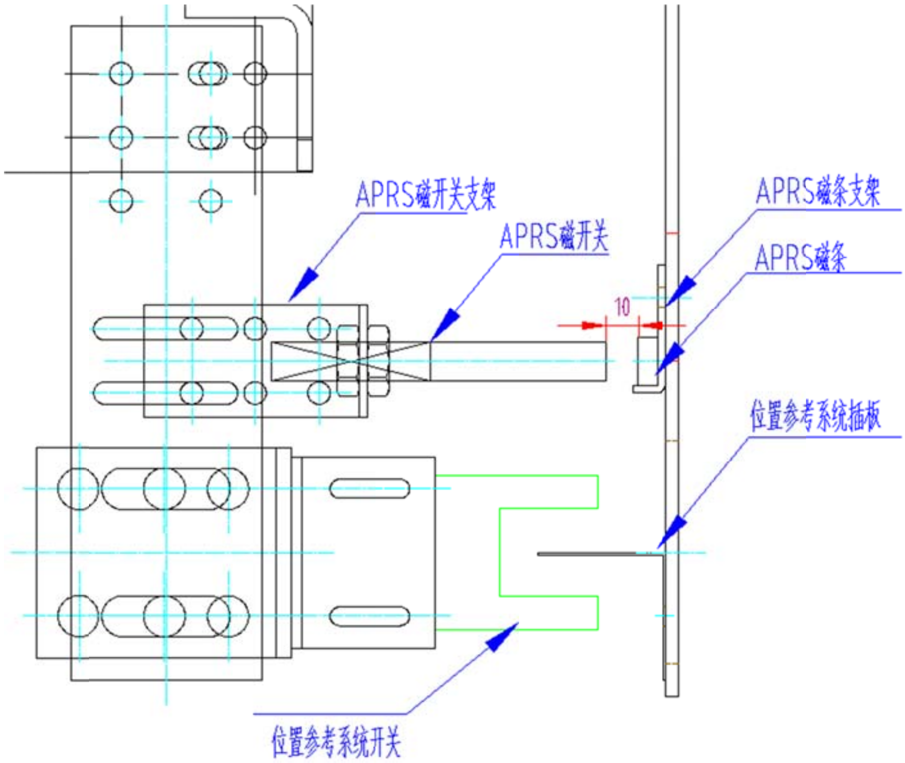
When the elevator has blind floor, a magnetic switch will be equipped to the original position based on the system PRS, and the switch together with the magnetic stripe installed in the well constitutes additional position reference system (APRS, Additional position reference system), so that when the elevator system is carrying out reset rescue operation, it can identify the blind floor and arrive at safe floor to release passengers.

2.16.1 APRS installation

When the elevator has a single door, if there is a blind floor inserting plate, then install a magnetic stripe at the inserting plate place. No need to install if there is no blind floor inserting plate;

When the elevator has double door, currently the system only supports the front door opening reset and rescue leveling function, if the front door has a blind floor inserting plate, then install magnetic stripe at the blind floor inserting plate location only, if the front door has no blind floor inserting plate, then no need to install the magnetic stripe.

Specific installation can refer to the following figure:



APRS installation diagram

Note:

1. APRS signal is uploaded through the serial communication, so when configuring APRS, you must configure a serial communication board at the same time;
2. ARPS is recommended to configure with magnetic stripe or photoelectric switch.

2.16.2 Function debugging

After the blind floor magnetic switch is confirmed to be installed, the debugging of the rescue reset operation can be further carried out.

Please set the corresponding parameters according to the table below:

Menu	Parameter name	Parameter value	Setting instructions
M-1-3-1-1	CR-OPT	0	CR-OPT is the reset rescue operation mode selection: +1: start the rescue operation function, when +8 is not selected, it indicates that the floor with landing door is installed with APRS; +2: stop to rescue at the location of the security door; +4: when the door is opened in the rescue floor, cancel the buzzer; +8: install APRS on the floor in absence of landing door;; Set to 13 indicates (1+4+8) to start the reset and rescue level function and cancel the buzzer when the door opens in rescue leveling, and install APRS at position with no landing door.
M-1-3-1-1	CR-DAR-T	15	Door opening and waiting time in rescue operation
M-1-3-1-3	CR-CHK-T	0	After the door lock contact is in action for 1s, confirm whether the landing door lock contact is in action
M-1-3-2	1778 CR-FSO	56-1	RSL address 56-1
	1779 CR-RSO	01-0	RSL address 01-0
	1775 GOL	62-2	RSL address 62-2
M-1-3-2	1776 MD_COR	62-3	RSL address 62-3
	1777 MD_BTN	62-4	RSL address 62-4
M-2-2-2	Reset At PowerOn	0	0: the motherboard loses power and gains power again, and the elevator maintains the state before losing power; 1: the motherboard loses power and again gains power, and the elevator goes to the terminal for reset

Note: in opening the rescue operation function, please confirm the above parameters are set in accordance with the requirements.

When setting the CR-OPT > 0, and the elevator has a blind floor (judge according to the M-1-3-3-1 Enable setting), the server status bar prompts "DCS Run", which means the system requires a DCS self-learning, to identify the location of APRS, otherwise this function cannot be opened.

If you want to have DCS learning, please enter the M1-3-5 menu for DCS door detection operation. After DCS learning is completed, press M-1-1-1 to check whether "DCS Run" prompt has disappeared, if it has disappeared, it indicates that the system has identified the blind floor position, the reset and rescue operation function can normally work, and you can continue the other debugging of the system. Otherwise, you need to check whether the parameter setting or APRS installation has any problem, until there is no prompt “DSC Run” after DCS learning.

2.17.2 Motherboard parameter settings

Take main engine YTTD160TVF2-4 of YJ200 for example, confirm the parameters as follows:

Menu key value	**Display content**	**Chinese explanation**	**Parameter setting value**	**Remarks**
M-2-2-1	Gear ratio	Reduction ratio	41	Contract parameter
	Rated RPM	Main engine rated rotate speed	1440	Contract parameter
	Shv diam (mm)	Traction wheel diameter	530	Contract parameter
M-2-2-2	Accelera normal	Acceleration	250	Set as 150-250
	Jerk0 normal	Jerk 0	250	Set as 150-250
	Jerk1 normal	Jerk 1	250	Set as 150-250
	Decelera normal	Deceleration	250	Set as 150-250
	Jerk2 normal	Jerk 2	250	Set as 150-250
	Jerk3 normal	Jerk 3	150	Set as 50-150
M-2-2-4	Inertia kg/m2	System inertia	2	Normally set below 10
	Start Kp	Start ratio regulator gain	0	Set as 0
	Start Ki	Start differential regulator gain	0	Set as 0
M-2-2-5	Encoder Sort	Encoder sort	0	1313 encoder set as 0
	Rated Power(Kw)	Main engine rated power	11	Contract parameter
	Number of poles	Main engine motor pole number	4	Contract parameter
	Rated frq (Hz)	Main engine rated frequency	50	Contract parameter
	Rated voltage(V)	Main engine rated voltage	380	Contract parameter
	Duty load (kg)	Main engine rated load	1000	Contract parameter
	Rated I (A)	Main engine rated current	23	Contract parameter
	Rated Trq (Nm)	Main engine rated torque	80	Set as 80
	Control methord	Main engine control method	1	Asynchronous machine is set as 1
	induct d0(mH)	d0 axis inductance	2.4	Set according to reference value

	induct q0(mH)	q0 axis inductance	2.4	Set according to reference value
	induct d(mH)	d axis inductance	2.4	Set according to reference value
	mutual resist	Mutual resistance	0.37	Set according to reference value
	induct q(mH)	Q axis inductance	2.4	Set according to reference value
	Rotor Time (s)	Rotor time	0.38	Set according to reference value
	No load current	No-load current	6.3	Set according to reference value

The following is the various types of asynchronous motor debugging parameters, for reference only.

Serial No.	M-2-2-4	M-2-2-5						
	Inertia kg/m2	**Rated Power(Kw)**	**mutual resist**	**No load current**	**Rotor Time (s)**	**Ld0, Ld Lq0, Lq**	**Rated Trq (Nm)**	**Number of poles**
1	2	6.4	0.803	3.5	0.32	4.7	80	4
2		7.5	0.597	5.15	0.351	2.9		
3		9	0.494	6.43	0.337	2.3		
4		11	0.367	8.43	0.333	1.7		
5		12.5	0.309	8.78	0.383	1.6		
6		15	0.267	11	0.4	1.4		
7		18.5	0.195	12	0.41	1		
8		22	0.155	14.5	0.41	1		
9		30	0.12	19	0.42	1		
10		37	0.1	21.5	0.45	1		
11		45	0.1	25	0.48	1		
12		7.5	0.579	10	0.21	3		6
13		11	0.362	13.3	0.22	2		
14		15	0.23	16.5	0.26	1.5		
15		18.5	0.193	17.5	0.3	1		

2.18 RS communication board instructions

RS communication board is commonly RS5 communication board and RS32 communication board.

2.18.1 RS5 communication board

2.18.1.1 Product physical map

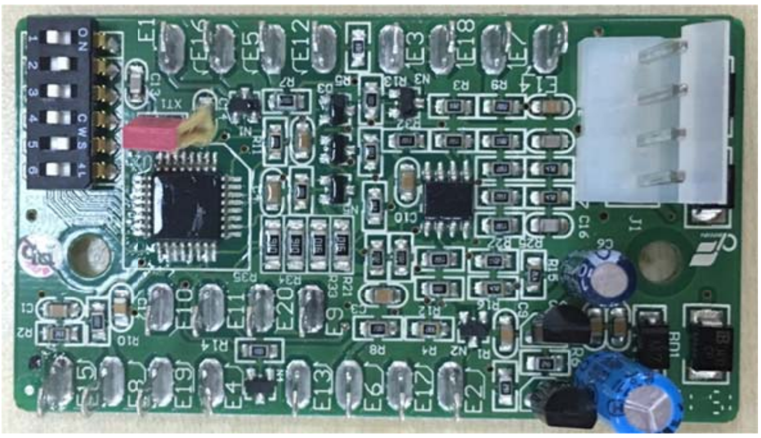


Table 2 Power supply and RSL communication signal interface instructions

2.18.1.3 Product features

This product is used for data transition between RSL bus and the elevator control system. Figure 3 is the typical access method of four set of input / output signal.

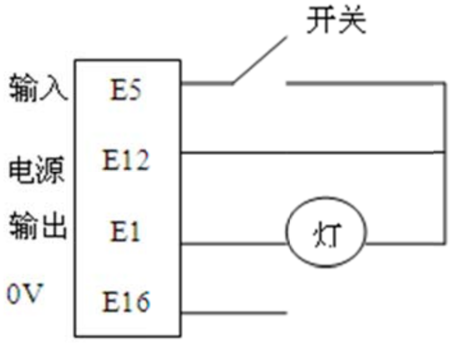


Fig. 3 Typical circuit connection

2.18.1.4.2 Preparation before installation

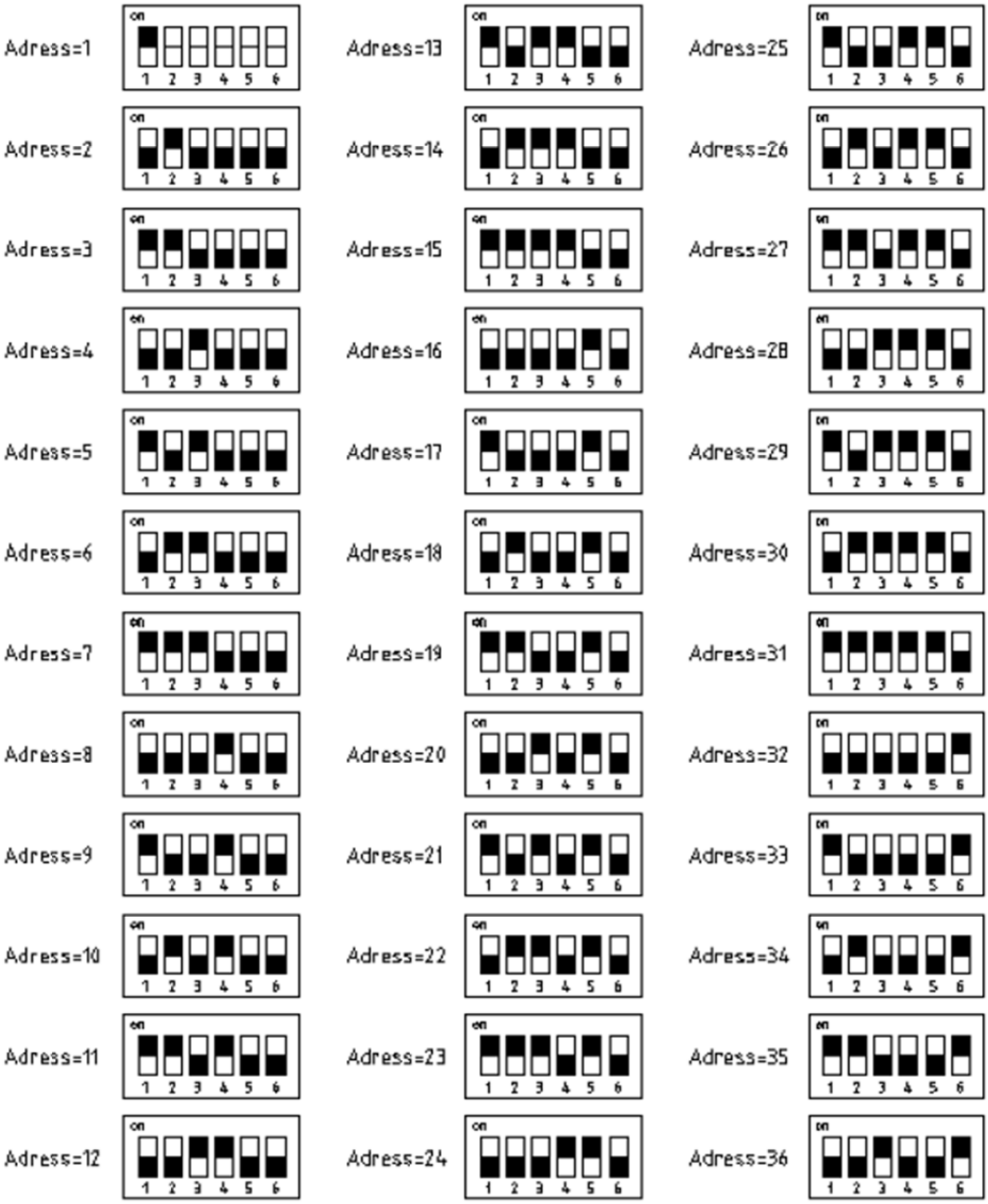
According to the site, adjust the correct RS5 communication board address code.

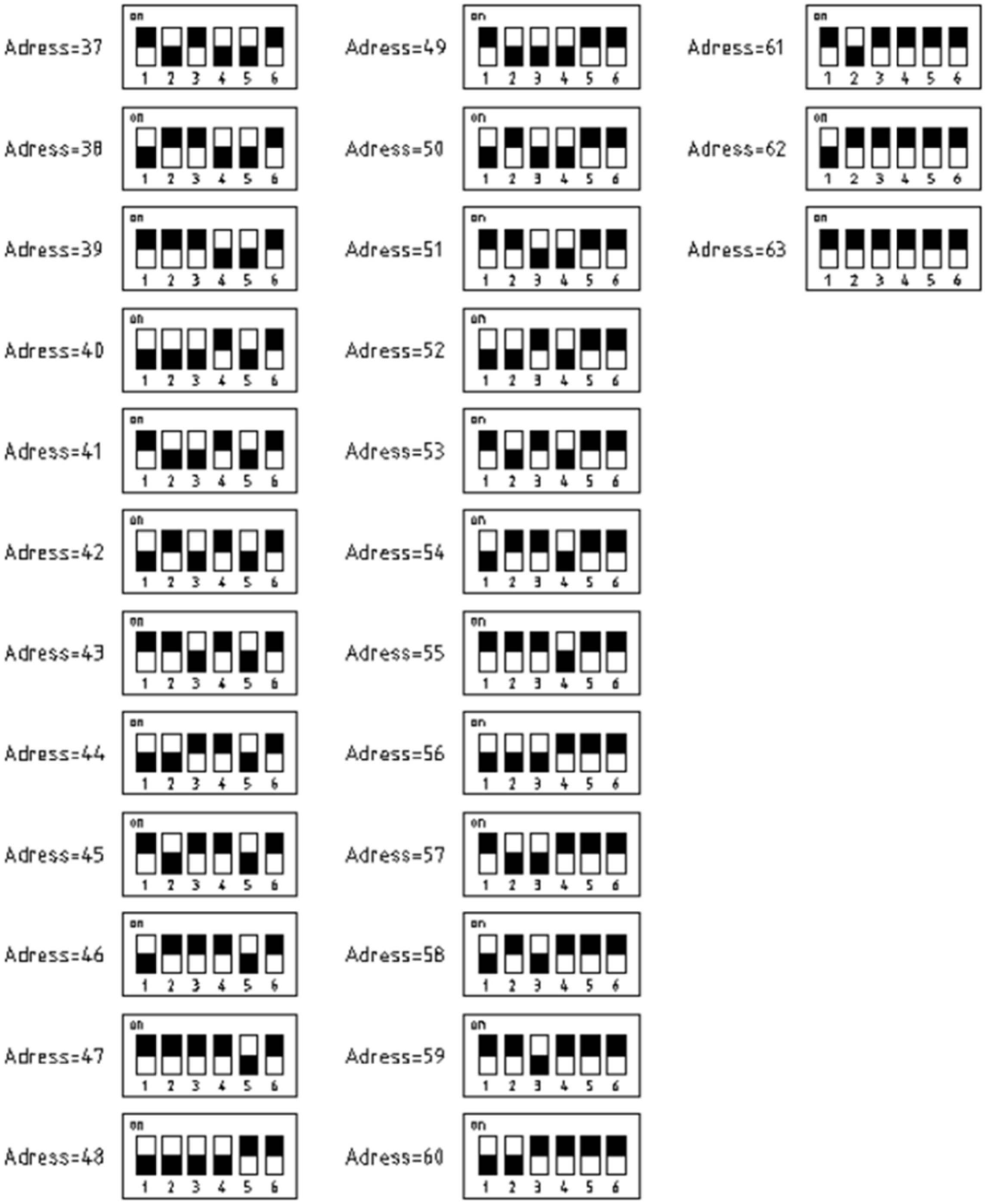
2.18.1.4.3 Installation and adjustment steps

By dialing the code switch you can set the address code of the site. Details are as follows (high 1 low 0):

$$0\times2^1+0\times2^2+1\times2^2+1\times2^3+1\times2^4+1\times2^5=60$$

According to the above description, turn the dial switch to different address. The state of each dial switch is shown in the following figure.


Adress=1
Adress=2
Adress=3
Adress=4
Adress=5
Adress=6
Adress=7
Adress=8
Adress=9
Adress=10
Adress=11
Adress=12
Adress=13
Adress=14
Adress=15
Adress=16
Adress=17
Adress=18
Adress=19
Adress=20
Adress=21
Adress=22
Adress=23
Adress=24
Adress=25
Adress=26
Adress=27
Adress=28
Adress=29
Adress=30
Adress=31
Adress=32
Adress=33
Adress=34
Adress=35
Adress=36



Note: When the dial switch address is 1 and 2, RS5 communication board is in invalid state

Fig.4 .6digit dial switch instructions

2.18.2 RS32 communication board

2.18.2.1 Product physical map

2.18.2.3 Debugging steps

Debugging takes the following steps:

a) Connect each connector according to the definition of the interface, details refer to the control cabinet electrical schematic;

b) After confirming the wiring is correct, power on the system, LED1 lights up, and according to the system address configuration, set the input and output address, pay attention the input address and other input address do not duplicate, and the output address and other address don't duplicate, if it does, the two ports are invalid; if the system configuration uses the default configuration (factory default address refers to 6.17.2.4.1), it is directly available; if it does not use the default configuration, please set according to the server instructions. (See 6.17.2.4.2)

2.18.2.4.2 Server instructions

a) First insert the server serial port head into the RS32 board, the server displays the version information, confirm the software version information (XOEC or XIZI);

b) Press the main menu <M>key to enter the password authentication, the factory password: 1234, and enter the main menu after input is normal;

c) Enter the port monitoring menu (M-1) , you can press GO ON and GO BACK key for port selection, and press〈M〉to return to the main menu;

d) d) Enter RSL monitoring menu (M-2) , you can press GO ON and GO BACK key to make ADR address selection of RSL, you can press UP and DOWN key to select BIT address of RSL, and press〈M〉to return to the main menu;

e) Enter the parameter setting menu (M-3) , you can choose to enter port setting and password setting, and press〈M〉to return to the main menu;

f) Enter the port setting (M-3-1) , you can press GO ON and GO BACK key for port selection, and press ON and OFF key for ADR address selection of RSL and, you can press UP and DOWN key for BIT address selection of RSL, after the completion of the setting, you need to press ENTRY key to save the corresponding port setting and write into EEPROM, otherwise the parameters will recover to the parameters before the modification when power down. (Note: to set a port address in program version V1.0, you must press ENTRY key one by one to complete the setting, and then set the other ports, otherwise, you can only change the last set port when power down; and in program version V1.1, after all port addresses are modified, press ENTRY key to complete the setting, otherwise, the parameters will recover to the parameters before the modification when power down) ;

g) Enter password setting (M-3-2) , enter old password and press ENTRY key, enter new password for the first time and press ENTRY key, and enter new password for the second time, and press ENTRY key to complete the password setting, and return to the main menu;

h) Enter initialization menu (M-3-3) of parameter address table, enter 1, and confirm whether it is the initialization menu, menu address is 8~11, and then enter 1, carry out initialization tasks; enter 2, to confirm whether it is initialization menu, menu address is 12~15, enter 2, to perform initialization task; enter 3 or 4, to confirm whether it is the implementation menu, the menu address is empty, enter 3 or 4 to perform initialization tasks;

Note: when the ODS is configured as 1 piece of RS32 board, use the initialization menu 1 of parameter address table, when the ODS is configured as 2 pieces of RS32 board, the first RS32 is configured as initialization menu 1, externally connected to door opening and closing signal; the second RS32 is configured as initialization menu 2, only connected to button signal.

i) Enter version information (M-4) , and press〈M〉to return to the main menu.

j) Enter the anti false trigger set (M- Shift+7), press 1+ENTER to open the anti false trigger function, the menu displays Misinput:On; press 0+ENTER to indicate closing the anti false trigger function and the menu displays Misinput:Off; pay attention to press ENTER key to save parameters, otherwise they will return to the original parameters after power down; press "M" to return to the main menu.

2.19 Door machine mode instructions

1. DT code door machine type

XIZI system can configure the DT code type door machine, and send through the RSL communication the door opening and closing signal to perform door opening and closing action.

Parameter name	Parameter menu	Parameter value	Parameter value meaning	Factory default value
DOOR	M1-3-1-5	14	Front door machine type	5
REAR		14	Rear door machine type	0

DT code status is as follows:

DT1	DT2	DT3	Status
1	1	1	Door opening
0	1	1	Leveling door closing
0	1	0	Non-leveling door closing
0	0	1	Maintenance state
1	0	0	Slow door closing
0	0	0	Free state
1	0	1	Reservation function
1	1	0	

2. DO/DC door machine type

Parameter name	Parameter menu	Parameter value	Parameter value meaning	Factory default value
DOOR	M1-3-1-5	5	Front door machine type	5
REAR		5	Rear door machine type	0

DO	DC	Status
1	0	Leveling door opening
1	0	Non-leveling door opening
0	1	Leveling door closing
0	1	Non-leveling door closing
0	0	Free state

Door machine itself tests to judge whether the door can be grilled.

2.20 Brake torque detection function

This function can be divided into manual brake torque detection and automatic brake torque detection.

1. Manual brake torque detection mode:

Menu M1-2-2 interface as follows:

```
Load = 0%
N=0 Y=1
```

The interface prompts to confirm whether the car is no-loaded, if it is not no-load state, enter 0 to exit the current interface and return to the main menu, if it is no-load state, enter 1 to enter the next level interface:

```
BrkTest:0 St:STP
Cur:0000.0 Rst:
```

Enter the Sn shortcut menu of onboard server, press CHC and DDO key (corresponding LED lights up), enter maintenance, and enter 1 into BrkTest value, enter S+ENTER key the brake torque detection starts.

2. Automatic brake torque detection mode:

According to the factory settings BrkT-D=1 (detection interval), BrkT-H=3 (detection time hour), BrkT-M=0 (detection time minute), 3 am every morning it automatically executes a brake torque detection.

UCMP test method:

I. When configured with advanced door opening or re-leveling function (with LVCT1 board)

UCMP test conditions: activate the advanced door opening or re-leveling function (DRIVE parameter value under M1314 menu is set as 1, EN-RLV parameter value under M1316 is set as 1)

1. ensure that no one is inside the elevator car and take proper protective measures outside the landing door on the test floor, and prohibit any personnel entering and leaving the car;
2. elevator normally parks inside the lock area, enter M1-2-7 through the server, and enter the password: 7588 to enter the following interface (after 25s the interface will automatically exit, you need to re-enter);

Descriptions on the above interface:

F: display 0 to indicate there is no fault, display 1 to indicate currently it reports UCM fault;

I: run direction;

**: current floor;

NR: elevator not run;

C: call the elevator;

1lv2lv: Leveling photoelectric signal state, 1lv, 2lv capital means there is entry;

dfcdw: dfc car door lock switch, dw landing door lock switch, dfc and dw capital indicates there is entry;

][: front door and rear door status, details see 5.3.1.

1. Call the current floor through the server and, when the door is opened or opened in place (F:0, 1LV, 2LV, DFC in uppercase, dw lowercase) call other floor through the server, press the shift+2 button of the server to make the elevator door open for operation;
2. The elevator runs out of the door area in door opening state, and the stop device action makes the elevator stop, F:1 in M1-2-7 interface and, through the server enter M1-2-1 to view the status of fault, and the elevator reports UCM fault;
3. The elevator control system is power down, measure the sliding distance of car accidental movement protection.

II. When there is no advanced door opening or re-leveling function (no LVCT1 board) ;

UCMP detection method, detect the brake torque according to the brake torque detection function, refer to Item 49 brake torque detection function.

UCMP reset operation mode:

1. Confirm the door lock loop connected;
2. Mode one: in maintenance or emergency electric state, press TOP and BOT simultaneously for 5s, and clear UCMPfault;

Mode two: in maintenance or emergency electric state, enter M1-2-7 through server, enter password: 7588, press Shift+0 key, and clear UCMP fault (this mode can be used when there is no engine room) ;

2.21 Elevator call run enable setting

Allowed ENABLE Masks (SVT Key Sequence M-1-3-3-1-GOON) elevator call run enable setting					
At level	**CUDE**	**CUDE**	**P**	**R**	**Remark**
0	1100	0000	0	0	Contract parameter
1	1110	0000	0	0	
2	1110	0000	0	0	
3	1110	0000	0	0	
4	1110	0000	0	0	
5	1110	0000	0	0	
6	1110	0000	0	0	
7	1110	0000	0	0	
8	1110	0000	0	0	
9	1110	0000	0	0	
10	1110	0000	0	0	
11	1110	0000	0	0	
12	1110	0000	0	0	
13	1110	0000	0	0	
14	1110	0000	0	0	
15	1110	0000	0	0	
16	1110	0000	0	0	
17	1110	0000	0	0	
18	1110	0000	0	0	
19	1110	0000	0	0	
20	1110	0000	0	0	
21	1110	0000	0	0	
22	1110	0000	0	0	
23	1110	0000	0	0	
24	1110	0000	0	0	
25	1110	0000	0	0	
26	1110	0000	0	0	
27	1110	0000	0	0	
28	1110	0000	0	0	
……	……	……	……	……	
53	1110	0000	0	0	
54	1010	0000	0	0	

2.22 Display setting

POSITION INDICATOR (SVT Key Sequence M-1-3-4-GOON) display setting				
At level	**PI Display**	**Left**	**Right**	**Remark**
0	“-1”	37	1	
1	“1”	10	1	

2	“2”	10	2	
3	“3”	10	3	
4	“4”	10	4	
5	“5“	10	5	
6	“6“	10	6	
7	“7“	10	7	
8	“8“	10	8	
9	“9“	10	9	
10	“10“	1	0	
11	“11“	1	1	
12	“12“	1	2	
13	“13“	1	3	
14	“14“	1	4	
15	“15“	1	5	
16	“16“	1	6	
17	“17“	1	7	
18	“18“	1	8	
19	“19“	1	9	
20	“20“	2	0	
21	“21“	2	1	
22	“22“	2	2	
23	“23“	2	3	
24	“24“	2	4	
25	“25“	2	5	
26	“26“	2	6	
27	“27“	2	7	
28	“28“	2	8	
29	“29“	2	9	
30	“30“	3	0	
31	“31“	3	1	
32	“A“	10	11	
33	“B“	10	12	
34	“C“	10	13	
35	“D“	10	14	
36	“E“	10	15	
37	“F“	10	16	
38	“G“	10	17	
39	“H“	10	18	
40	“I“	10	19	
41	“J“	10	20	
42	“K“	10	21	
43	“L“	10	22	
44	“M“	10	23	
45	“N“	10	24	

46	“O“	10	25	
47	“P“	10	26	
48	“Q“	10	27	
49	“R“	10	28	
50	“S“	10	29	
51	“T“	10	30	
52	“U“	10	31	
53	“V“	10	32	
54	“W“	10	33	
55	“X“	10	34	
56	“Y“	10	35	
57	“Z“	10	36	

Common faults and troubleshooting methods

8.1 Common logic fault table

Access to M1-2 to view logic fault, turn to the current logic fault and press 1+2 to remove the current fault, keep pressing 1 and then press 2 to continuously delete logic fault.

Event	Possible cause	Relevant setting
total runs	Total running times after power on for the last time	-
minutes on	Minutes of running after power on for the last time	-
Flashing information		
Event	**Possible cause**	**Relevant setting**
LS - Fault	Forced upper deceleration 2LS and forced lower deceleration 1LS signal is not normal	
1LS + 2LS on	Forced deceleration switch 1LS and 2LS action at the same time	-
DBSS - Fault	Inverter is not ready	-
TCI - Lock	Car top maintenance switch operation sequence is not accurate, it is necessary to open the landing door according to the following sequence. 2 car top maintenance switch turns from the maintenance to the normal condition.3. close the landing door If you do not operate in accordance with the above sequence, the elevator will not operate normally, while the INS indicator on logic control board will blink.	-
DBP - Fault	The elevator must run in the right order. The signal of door area has not been detected when the elevator slows down and stops. The reason may be the fault of the LVC relay. This error is stored in the EEPROM of logic control part, and you can only use the INS maintenance switch to operate the motor.	
Adr - Check	The address of the special remote station does not correspond to the standard IO list.	

start DCS!	a) The normal operation is not allowed before the door check start (DCS) operation is completed. b) In normal operation, when the elevator stops in the door area and the car door is fully opened (DOL valid), the door closing signal is triggered (DW valid). At this point, please carefully check whether the well landing door lock is short circuited. Press M - 1 - 3 - 5 to start DCS!	NoDW - Chk
UCM-Lock!	Car accidental movement triggers fault **Note: if you need to clear the fault, in the maintenance state press the motherboard BOT and TOP key simultaneously for 3 seconds, the fault will be cleared.**	
Event	**Possible cause**	**Relevant setting**
DLM	In the process of elevator opening and closing, the following circumstances may trigger DLM: 1. DW is short circuited; 2 DFC is short circuited; 3 when DFC and DW are short circuited, enter the DLM mode, try to open and close the door 3 times, and the door opening waiting time refers to the normal door opening waiting time. All call instructions remain in this process, but there is no response. If the signal is detected not to short circuit at the third time door opening, then exit DLM mode. If it detects short circuited at the third time, then it flashes "Door Bridge". At this point, all call instructions are cancelled, and no other instructions are registered. If it is the fourth time to release passengers, the door opening waiting time shall be set in accordance with the DAR-T. After the door is closed and DOB and DOS in the car are valid, then the door can be opened, and closed after the DAR-T time. EDP and LRD can effectively open the door in the closing process. At this time through the ERO/TCI OFF->ON->OFF, the elevator opens the door to re check the door lock state, if normal, then the elevator returns to normal mode. If not normal (if still short circuited), wait for DAR-T time and then close.	

	Power off and re start, DLM mode reserved Matters needing attention: 1, In detection of signal, within three times, as long as it detects the door lock is disconnected once, then immediately return to normal; 2, After the fault is triggered, you can only use the "ERO/TCI action" to remove, power on and power off is invalid;	
HAD	In normal mode, nearest leveling and, reset operation, when the car door is closed, and the landing door opens more than 4S, the logic control system reports HAD. After HAD is triggered, if the landing door is closed, the elevator will run to the next floor (non-blind floor) at rescue speed (300mm/s), if it is blind floor, the elevator continues to run to the next floor, until stops at the normal floor. Stop and open the door, and wait for DAR-T time to close. And in the process the registered instructions inside the elevator / landing hall will be cleared, and no longer responds to any registered instructions. After the door is closed and DOB and DOS inside the car is valid, the door can be opened, and closed after the DAR-T time. EDP and LRD can effectively open the door in the closing process. Matters needing attention: 1, After the fault is triggered, you can only use the "ERO/TCI/ESB action for over 2S" to clear and, power on and power off is invalid; 2, If the elevator is reset to the end station, it will continue to leave, and will stop at the floor nearest to the place of forced deceleration; 3, Within 4 seconds after the landing door is disconnected, press the emergency stop button (bottom pit, car top, control cabinet), it will not trigger the HAD mode.	
OCSS		
Event	**Possible cause**	**Relevant setting**
0100 OpMode NAV	Drive part fault causes OCSS not to be operated, and this model can also be triggered after maintenance and before correction.	-
0101 EPO shutd.	Car can’t be run in emergency power operation mode EPO.	NU (017) NUSD (018)

		NUSG(019)
0102 OpMode DTC	The door can't be closed normally (lost DCL, DFC or DW) within the set time.	DCL (1206) RDCL (1207) DOOR, REAR DTC-T
0103 OpMode DTO	The door can't be normally opened in place within the set time	DOL (000) RDOL (1056) DOOR, REAR DTO-T
0104 OpMode DCP	The car can't respond to a call or instruction (for example, the door is blocked by obstacles) within a set time.	DCP-T
0105 DBSS fault	Driver fault	DRIVE
0106 PDS active	Part of the door lock switch not closed	PDD (1296)
MCSS		
Event	**Possible cause**	**Relevant setting**
0200 Pos. Count.	After the operation is completed, the system detects the door area and the IP signal count does not match, it may be the event issued by door area signal DZ and IP signal is too short, the system does not have the time to detect.	LV-MOD, DZ-TYP
0201 correct. run	Correct operation (for example, after power on, maintenance run, and NAV, etc.).	-
0202 /ES in FR	ES signal is activated when the elevator runs at fast speed	MD/AES, ES-TYP
0203 /ES in SR	ES signal is activated when the elevator runs at low speed	MD/AES, ES-TYP
0204 TCI/ERO on	TCI or ERO switch subject to action	ERO-TYP
0205 SE-Fault	Elevator can't start due to the loss of SEsignal(check SKL, THB, fuses etc.)	-
0207 DDP in FR	When elevator is running at fast speed within set time (DDP) , no well signal is detected (DZ lost)	DDP
0208 DDP in SR	When elevator is running at low speed within set time (3P) , no well signal is detected (DZ lost)	3P
0209 DDP in RS	When elevator is running at rescue speed within set time (3P) , no well signal is detected (DZ lost)	3P
0210 /DZ in NST	No DZ signal is detected when elevator stops	LV-MOD, DZ-TYP
0211 /DFC in FR	Safety loop is disconnected when elevator runs at fast speed	-
Event	**Possible cause**	**Relevant setting**
0212 /DFC in SR	Safety loop is disconnected when elevator runs at low speed	-

0216 DZ missed	UIS and DIS signal is received but DZ signal not received, which may be caused by LV relay fault	EN-RLV, DRIVE
0224 J-Relay	The logic control part detects a fault in three-phase power supply (e.g., missing or faulty phase)	EN-J, J-T
0226 LS-fault	Forced deceleration signal is not normal, flashing information is seen.	-
0228 1LS+2LS on	Detect 1LS and 2LS signal at the same time	1LS (1204) 2LS (1205) DRIVE C-TYPE
0230 RSL Adr chk	See flashing information description	NoAdrChk
0231 LSVF-W:/DR	Inverter failure (not ready)	DRIVE C-TYPE *(see MCB)*
0232 LSVF-W:/SC	In deceleration the elevator speed is too high, and it can't complete the advanced door opening function ADO or re leveling function RLV.	*(see MCB)*
0237 /DW in FR	When elevator is fast running the landing door circuit disconnected.	-
0238 /DW in SR	When elevator is slowly running the landing door circuit disconnected.	-
DCSS		
Event	**Possible cause**	**Relevant setting**
0300 DBP: dfc_SE	When the door is opening or the door has been completely opened, DFC or SE (with ADO function) has no action	EN-RLV DRIVE
0301 DCL in [ ]	When the door is completely opened, detect DCL signal	DCL (1206) RDCL (1207)
0302 DCS:DW err	During normal operation, when the door is opened, detect the landing door circuit closed	NoDW-Chk
0303 DBP-Fault	See flashing signal 'DBP-Fault'	-
0304 DOL:alw. on	When the door is completely closed, detect the DOL signal; if the DO2000 door machine fuse is broken, the fault will also occur.	DOL (000) RDOL (1056)
SSS		
Event	**Possible cause**	**Relevant setting**
0400 RSL parity	Two remote stations connected to the same serial line use the same address.	-
0401 RSL sync	Synchronous signal on the remote serial line is lost.	-
GROUP		
Event	**Possible cause**	**Relevant setting**

0500 RNG1 msg	Data in parallel / group control serial loop has error.	-
0501 RNG1 time	There is no signal received from other elevators during a certain period of time.	GROUP
0502 RNG1 sio	Transmission format of parallel / group control serial circuit has fault.	-
0503 RNG1 tx	Serial data transfer timeout.	-

8.2 Common drive error table

Enter the M2-3 to view the current drive error, press the ON GO key to view the time and floor the error occurs, press the SHIFT+ENTER key to clear the current drive error

Fault code	Fault display	Fault name	Level	Fault system level trigger condition	Possible cause
ERR-01	IGBT Fault	IGBT/IPM module fault	B	Run in case of module over-current, output short-circuit, and brake not opened, the trigger signal is from the IPM module	A, Drive output side U, V and W has short circuit; B, Elevator brake is not opened while drive has already current output; C, When elevator is in normal maintenance, self learning and correction running, suddenly the door lock or safety loop is broken; D, IGBT module is damaged; E, Drive and motherboard wiring not in good contact
ERR-02	Over current	Drive over current fault	B	Run in case of frequency converter over current, output short circuit or brake not opened; or load is too big and current is too large, acceleration and deceleration too fast, drive size is too small, module damaged, encoder damaged. Trigger signal is from the current detection path, Holzer sensor.	A, drive output side V, U and W has short circuit; B, elevator brake is not opened but the drive has current output; C, when the elevator is in normal maintenance, self learning and correct operation, the door lock or safety loop broken; D, IGBT module damaged; E, cable damaged; F, positioning angle not correct; G, motor runs with phase missing J, encoder runs with signal lost

ERR-04	Overtemp	Radiator over-temperature	C	Module is overheated, any of the two temperature detection points, including the IPM and rectifier bridge is >85℃	A, temperature sensor damaged; B, drive cooling fan damaged; C, actual cooling fin reaches this temperature;;
ERR-06	Drive overload	Drive overload	B	When the control board CPU uses the integral points of output current, the totalpoint exceeds a certain value, it triggers drive overload fault.	A, brake not opened in the operation of elevator; B, positioning angle of main engine not correct; C, overload does not work and actual elevator overloaded; D, motor protection coefficient setting is too small
Fault code	**Fault display**	**Fault name**	**Level**	**Fault system level trigger condition**	**Possible cause**
ERR-07	DC link OVT	DC bus overvoltage	C	1, When the control panel CPU detectsthe overvoltage signal of hardware, it will trigger overvoltage fault. 2, When the control panel CPU detects the bus voltage exceeds the software overvoltage point set value, it will trigger overvoltage fault. DC bus voltage is overvoltage	A, brake resistance short circuited; B, external input voltage too high; C, control board first power on and then the drive power on, due to the impact it may cause overvoltage fault
ERR-08	DC link UVT	DC bus under-voltage	C	1, When the control panel CPU detects the under-voltage signal of hardware, it will trigger under-voltage fault. 2, When the control panel CPU detects the bus voltage exceeds the software under-voltage point set value, it will trigger under-voltage fault. DC bus voltage is overvoltage	A, input RST power voltage below 280V; B, no power on the base; C, parameter “Bus fscale (V) ”not correctly set; D, rectifier bridge or soft start resistor fault;;

ERR-09	Overspeed	Main engine over-speed	B	Main engine running speed exceeds rated speed (M3312) times over speed	A, fast run speed “Velocity normal ”is set too big; B, inertia “Inertia kg/m2” is set too big; C, encoder related parameters, traction wheel diameter, reduction speed ratio and rope winding ratio parameters are not set correctly; D, frequency converter has no output;
ERR-10	PVT lost	Encoder fault	B	Encoder signal is lost, abnormal or encoder hardware fault (such as power supply problem or broken line etc.) the drive does not detect the encoder signal	A, encoder line not connected or broken; B, motor phase is not correct in positioning or the direction of encoder not correctly set; C, encoder pulse number and type is not correctly set D, encoder line and shield line is short circuited
Fault code	**Fault display**	**Fault name**	**Level**	**Fault system level trigger condition**	**Possible cause**
ERR-12	DRV Comm timeout	Communication error	C	Not received drive communication data or data incorrect	A, motherboard and the drive cable is not connected or the connection is not reliable; B, motherboard or driver hardware communication circuit has abnormalities;
ERR-13	Task orun	Pwm interrupt task execution overtime	B	1ms, 10ms, and 40ms interrupt task execution time is overtime, and an interrupt cycle does not fully execute all of the above tasks	A, PWM carrier frequency "FRQ Switch (KHz)" is set too large;
ERR-14	Tune Moving	Encoder angle self learning engine moves	B	In the self - learning pole and the encoder angle, the rotor moves, leading to the failure of self - Learning the movement of motor is too large during motor auto phasing	A, brake clearance is too large, causing the main engine not fully embraced; B, encoder is not completely fixed;;

Fault code	Fault display	Fault name	Level	Fault system level trigger condition	Possible cause
ERR-15	Track error	Speed track error	B	When the actual motor detection speed of control panel CPU is compared with the set speed, and meets at the same time the following 2 conditions, it will trigger speed track error. ① abs (actual elevator speed－speed given) >elevator rated speed×Track error/100 ② Duration exceeds the set time value the gap between actual speed and command speed is more than "Track err"	A, brake is not turned on in elevator operation; B, main engine positioning angle is not correct; C, encoder does not move when main engine runs; D, encoder signal is not correct; E, Track error value is too small and may occur in maintenance operation;
ERR-16	1LV NORMAL CLOSE	1LV photoelectric fault	A	Elevator runs 1000mm 1LV photoelectric signal keeps valid	Leveling sensor switch invalid
ERR-17	2LV NORMAL CLOSE	2LV photoelectric fault	A	Elevator runs 1000mm 2LV photoelectric signal keeps valid	Leveling sensor switch invalid
ERR-18	Floor number err	Floor number fault	A	Elevator self learning is completed and with 2LS, the floor learned is not equal to the set floor height	Floor parameter setting error
ERR-19	PARA. ABNORMAL	Parameter setting fault	A	The set motor frequency, motor speed and the pole number does not meet the formula N=60F/P	A, Main engine parameter setting error
ERR-20	Sfty chain state	Safety loop fault	C	DFC is detected to disconnect in the operation	A. external DFC signal is abnormal in the operation B, motherboard or drive DFC circuit abnormal

ERR-25	E2 write err	Eeprom reading and writing error	B	Control panel CPU will perform EEPROM reading and writing in power-on self check and, according to the special unit reading and writing it can determine whether the EEPROM can be normally operated, if reading and writing data are inconsistent for 3times, then it will trigger fault. In the normal operation after the completion of power on, if EEPROM shall be written as required, first write and then re-read, consistency is normal, and if it is inconsistent do it for 3 consecutive times, if not consistent, it will trigger fault. E2 write err	A, EEPROM chip damaged;

Fault code	Fault display	Fault name	Level	Fault system level trigger condition	Possible cause
ERR-28	Brake dropped	Brake switch 1 status fault	B	The control panel CPU continuously detects the brake switch status, and compare the brake instruction status of software command with the actually detected brake switch status, if the continuous detected inconsistent state exceeds the brake switch detection time, it will trigger the fault. Including the following2circumstances: ① can't be opened in	A, actual action of brake switch does not meet the fault design requirements; B, wrong wiring; C, brake switch detection time is set too small (default 500ms) ;
ERR-35	Brake dropped 2	Brake switch 2 status fault	B		

				operation process; ② can't be closed when it stops; The brake is not opened or closed completely	
ERR-29	AC Line imbal	Output three-phase imbalanced	B	Output three-phase imbalanced	A. Drive damaged B. Relative short circuit
ERR-30	AC Line UVT	Three-phase input under-voltage	B	430 not used in first time power on	A, Drive damaged
ERR-32	Start Relay Err	Start relay error	B	Drive start relay not connected to the bus	A, Drive damaged B, External power under-voltage
ERR-33	DBD PICKUP	Main contactor or brake contactor has action fault	B1	Control panel CPU continuously detects the DBD input status and compare the contactor control of software instructions with brake control command status and DBD input status actually detected, if the continuously detected status inconsistency exceeds 100ms, it will trigger fault. Status consistency is judged as follows: ① When any of UDX and LB has output, enter 0 into DBD; ② when UDX and LB has no output, enter 1 into DBD; Maximum fault times is set in MAX DBD ERROR.	A, the actual status of LB or UDX not meet the design requirements B, motherboard power (DC24V) on first, then DC24Vpower on; C, in main engine positioning, short circuit UDX and LB, but set RUN SOURCE to be 1 (automatic) ;
Fault code	**Fault display**	**Fault name**	**Level**	**Fault system level trigger condition**	**Possible cause**

ERR-34	DDP Fault	Photoelectric disconnection	A	Control panel CPU will detect door area signal in normal operation and reset operation, when there is no door area pulse signal within the set time, it will trigger fault.	A, photoelectric switch is damaged, and no signal input; B, DDP time is set too small, this fault may occur in reset operation (half the rated speed) ,(in this case, move the forced deceleration switch to the two ends, or shield DDP, and set DDP time as 0);
ERR-36	Brk check Err	Braking force check error	A	In braking force detection, the main engine replacement exceeds the threshold with torque current given	Braking force is seriously insufficient, and need to replace the brake
ERR-39	Bus overcurrent	Bus over-current	A	Monitoring bus current exceeds threshold	A, output phase has short circuit; B, optocoupler fault of drive board hardware; C, output U, V and W has grounding short circuit; D, P and N has grounding short circuit;
ERR-42	InvIuOffst	U-phase current sampling error	B	When there is no current output, DSP □ U-phase current sampling value exceeds threshold	A, drive U phase current sampling channel is damaged, including Holzer, and operational amplifier circuit;
ERR-43	Inv IvOffst	V-phase current sampling error	B	When there is no current output, DSP □ V-phase current sampling value exceeds threshold	A, drive V phase current sampling channel is damaged, including Holzer, and operational amplifier circuit;
ERR-44	InvIwOffst	W-phase current sampling error	B	When there is no current output, DSP □ W-phase current sampling value exceeds threshold	A, drive W phase current sampling channel is damaged, including Holzer, and operational amplifier circuit;
ERR-47	Base AD Offst	Reference voltage	B	the 0.5V, 1.5V reference voltage deviation of driver board or drive base is too large	A, drive is damaged;
ERR-48	ETSC relay fault	ETSC fault	B	ETSC signal feedback error	A、ETSC contactor damaged; B、ETSC line problem; C、 ETSC parameter setting error;

Fault code	Fault display	Fault name	Level	Fault system level trigger condition	Possible cause
ERR-50	ctrler comm. Err	Drive communication fault	C	Drive communication data fault	A, Plug-pull communication line; B, Motherboard and drive cable not connected or the connection is unreliable; C, Drive and motherboard cable not in good contact
ERR-51	Position Lost	Location lost	C	In the running process the number of running floors and photoelectric number detected does not match, or the difference between elevator photoelectric position and self learning position is greater than 200mm	A, photoelectric fault; B, wire rope slips when elevator is running;
ERR-61	Vcode abnormal1	V code fault 1	C	The curve does not respond to the V code sent by logic, for 3s	Run source SVT=0 RUN ENABLE =0 EPC countdown is not completed DFC disconnected DBD feedback error
ERR-62	Vcode abnormal2	V code fault 2	C	The elevator stops and not respond to the V code sent by the logic for more than 100*"VS abnorm t (s)" ms	A, Lock is broken at low speed running; B, No safety circuit board, but set the advanced door opening function;
ERR-63	Power lost	Main power lost	C	After the power is lost, the bus voltage will be reduced, when it is less than DC380v, it will trigger the fault, when the power supply increases and bus voltage exceeds DC400v, the fault will automatically disappear.	A, normal power off/power on B, control panel has power and drive no power C, cable not plugged in
ERR-70	parm. dectBrk	Parameter association error	C	ARM&DSP brake switch detection parameter association error	Motherboard or drive has been changed but parameters are not reset
ERR-71	parm. 0spd	Parameter association error	C	ARM&DSP 0speed maintaining time parameter association	Motherboard or drive has been changed but parameters are not reset

				error	
ERR-73	parm. encode	Parameter association error	C	ARM&DSP encoder line number parameter association error	Motherboard or drive has been changed but parameters are not reset
ERR-74	parm. RPM	Parameter association error	C	ARM&DSP rated rotate speed parameter association error	Motherboard or drive has been changed but parameters are not reset
ERR-75	parm. DIAM	Parameter association error	C	ARM&DSP main engine diameter parameter association error	Motherboard or drive has been changed but parameters are not reset
ERR-76	parm. Gear	Parameter association error	C	ARM&DSP reduction ratio parameter association error	Motherboard or drive has been changed but parameters are not reset
ERR-77	parm. Rope	Parameter association error	C	ARM&DSP rope winding ratio parameter association error	Motherboard or drive has been changed but parameters are not reset
ERR-80	Custom code Err	Customer code error	C	Motherboard and drive not match	Motherboard or drive has been changed

8.3 Integrated ARD related fault

Serial No.	Fault	Fault cause level investigation
1	Power grid is power down, UPS started, but the system does not enter the EPR mode	1, Parameter setting problem, the motherboard IO 17 (NU: Emergency power) value is set to logic bit =0, address =1, bit =0 2, Hardware circuit problem, at this time the motherboard P1-7 should have DC24V voltage. If there is no voltage, power lost failure will be reported, and you need to check the PW contactor circuit and switching power supply line 3, Ensure all air switches in the control cabinet are in open state, otherwise it is unable to enter the ARD rescue
2	Power grid is power down and UPS not started	1, UPS start switch is not turned on 2, UPS battery power is not enough or damaged 3, Check whether UPS input terminal has AC220V, if input power is not cut off, UPS will not enter battery output mode
3	Power grid is power down and ARD rescure not started	1, UPSC contact is damaged and the coil does not work 2, UPSC contact is closed and, check whether UPSC opening contact 3, 4, 5, 6 is closed, otherwise the drive will have no power

4	Power grid is power down and motherboard not work	1, UPSC contact is damaged and the coil does not work 2, UPSC contact is closed and, check whether the opening contact 53, 54, 63, 64 is closed, otherwise the switch power will have no power supply causing the motherboard to have no power
5	ARD rescue direction and the actual car load direction is not consistent	1, If the car side is heavy, ARD direction is downward, and vice versa. If the direct is wrong, change the parameter ARD Run Direct=0/1 for correction 2, Ensure the encoder connection is correct, and UVW sequence is correct, and can meet the normal fast running conditions
6	ARD rescue leveling is not good	1, ARD speed is set too big, parameter range is 1-500, and recommended range 80-120 2, When it is double photoelectric, photoelectric distance installation is not in place, it is recommended that the two photoelectric spacing shall be 60mm 3, Photoelectric signal has abnormalities, and need to check the photoelectric signal wiring
7	ARD emergency stop in rescue process	1, The safety circuit is disconnected or the door lock circuit is broken, and the circuit is required to be recovered. 2, Control cabinet or car top has maintenance, maintenance recovery is normal, and the elevator continues ARD rescue 3, UPS power is not enough, need to ensure adequate charging for the first use 4, UPS type selection is not correct. When the main engine power is ≤11.7kw, and brake power is ≤ 250W without brake strong excitation function, UPS capacity shall be 1KVA/800W, model C1K 1KVA/800W; main engine power 11.7kw-25kw and brake power is ≤500W, UPS capacity is 2KVA/1600W, model C2K 2KVA/1600W 5, Report drive fault, the server monitors the fault menu, and make investigation according to the type of fault
Serial No.	**Fault**	**Fault cause level investigation**
8	ARD rescue to the leveling and door not opened	1, The motherboard DDO is switched to the off position, it should be to on. 2, Door opening and closing signal wiring error, according to the principle diagram check whether the wiring harness is correct 3, Detect the virtual floor signal, and need to check the virtual floor signal settings 4, EPO-DC parameter setting value is too small, this parameter is the set time when the door is closed after the rescue door opening is in place. Set the parameter according to the actual need

9	ARD rescue to the leveling and door not opened in place	1, Door opening in-place signal is not correct, need to check the connection and the relevant IO parameter settings 2, Detect the wrong rear door signal, and the rear door is opened incorrectly, need to check the location of rear door magnetic strip signal 3, Door machine got stuck , check the door machine machinery
10	ARD emergency stop in rescue process and start rescue again	1, Control cabinet or car top is marked as maintenance, and change maintenance to normal, the elevator continues ARD rescue 2, In ARD rescue process, after power on, the power grid is power down immediately, still take ARD rescue operation
11	Power grid gains the powr and ARD unable to change to normal mode	1, PW contact is damaged and drive no power 2. Check whether PW contact 71 and 72 opening point is disconnected, it shall be disconnected in normal power supply

8.4 Reset rescue function related fault

Serial No.	Fault	Fault cause level investigation
1	Elevator position lost , does not have rescue operation, and direct reset	1, Parameter setting problem, check whether the parameter CR-OPT is set to be rescue open;; 2, Magnetic signal error, check stripe installation and magnetic stripe signal parameter setting, and monitor whether the signal transmission is reliable; 3, The elevator rescues to the blind floor for three successive times, and fails to open the door to rescue, directly goes to the terminal station for reset 4, Report DCS run fault, troubleshooting refers to the following DCS run fault trigger 5, Report CR-FSO Flt fault, troubleshooting refers to the following CR-FSO Flt fault trigger
2	Report DCS run fault	1, Set the normal floor as blind floor, or change the blind floor to normal floor, the fault will be reported. Need to have DCS learning again 2, When the elevator normally runs and stops, detect the magnetic stripe signal, this fault will be reported. Check whether the elevator is running on staggered floor or magnetic stripe installed wrong, after rectification, need to learn DCS again 3, The elevator rescue operation stops at the floor absent of APRS signal and, when the door is opened to release passengers, the door contact is not disconnected, it will immediately close the door to reset at terminal station, and report the fault. Need to check whether magnetic stripe on each floor is installed according to the actual blind floor. If it is because the delay of landing door lock off is longer than the car door lock off, you need to adjust the DAR-T parameters,

		otherwise it will falsely report fault.
3	Report CR-FSO Flt fault	1, IO of APRS must be set with an effective address, if not, fault will be reported and the rescue function will be shielded; 2, Magnetic stripe signal trigger is detected when the elevator normally runs and stops, and this fault will be reported. Check whether the elevator is running on staggered floor or magnetic stripe is installed wrong, after rectification, need to learn DCS again; 3, The elevator rescue operation stops at the floor absent of APRS signal and, when the door is opened to release passengers, the door contact is not disconnected, it will immediately close the door to reset at terminal station, and report the fault. Need to check whether magnetic stripe on each floor is installed according to the actual blind floor. If it is because the delay of landing door lock off is longer than the car door lock off, you need to adjust the DAR-T parameters, otherwise it will falsely report fault.
Serial No.	**Fault**	**Fault cause level investigation**
4	No voice appease in the rescue process	1, check the voice appease related IO address settings and related wiring; 2, voice appease device damaged, you can replace the new test.
5	Rescue to the door area and door not opened	1. motherboard DDO switches to off position, it shall be to on position; 2, rescue to the floor and no APRS signal, but this is blind floor, and the elevator does not open the door, go immediately to the terminal station for reset; 3, door signal error, please check the door opening in place signal and door closing in place signal is normal or not
6	Elevator not reset after rescue is completed	1, safety circuit or door lock circuit is disconnected, please check related circuit ; 2, when the rescue operation direction has strong reducing action, at this point the position is corrected to be the terminal station, actually it has staggered floor, need to detect position error in next run
7	Emergency stop in rescue process	1, safety circuit or door lock circuit is disconnected and the circuit is required to be recovered; 2, photoelectric signal is lost, in the rescue process DDP failure is reported, need to check the photoelectric signal; 3, power down in rescue process; 4, report drive fault, the server monitors the fault menu, and

		make investigation according to the fault type
8	Rescue to leveling and door not opened in place	1, door opening in place signal is incorrect, need to check thewiring and relevant IO parameter settings; 2, door machine got stuck, check the door machine machinery
9	After emergency stop in rescue process, start rescue again	1, control cabinet or car top is marked as maintenance, recover maintenance to normal, the elevator continues rescue operation; 2, in rescue process the power grid is power on and power down immediately, still perform rescue operation; 3, safety circuit is disconnected in rescue process and continue to rescue after recovered to normal

8.5 Other system fault

Running status fault			
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
1	Elevator has emergency stop in fast running process and, not on the leveling, can't continue to run	1, safety signal is interrupted in operation; 2, door lock signal is interrupted in operation; 3, system has fault in operation;	1.ES on the motherboard lights up, use the server to view M112, ES signal in capital means safety circuit is disconnect 2. DFC or DW on the motherboard lights off, use the server to view the M112, dfc signal is in lower case and dw signal in lower case 3.ERR fault lamp is often bright, then view the M23 fault code, if ERR fault lamp is not bright, you can view the M23 fault code or M121 logic fault
2	Elevator has emergency stop in the maintenance operation and can't continue to run	1, safety signal is interrupted in operation; 2, door lock signal is interrupted in operation; 3, system has fault in operation;	1. ES on the motherboard lights up, use the server to view M112, ES signal in capital means safety circuit is disconnect 2. DFC or DW on the motherboard lights off, use the server to view the M112, dfc signal is in lower case and dw signal in lower case 3.ERR fault lamp is often bright, then view the M23 fault code, if ERR fault lamp is not bright, you can view the M23 fault code or M121 logic fault
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**

3	Elevator has emergency stop in the well self learning operation	1, safety signal is interrupted in operation; 2, door lock signal is interrupted in operation; 3, system has fault in operation; 4, floor number setting error;	1.ES on the motherboard lights up, use the server to view M112, ES signal in capital means safety circuit is disconnect 2. DFC or DW on the motherboard lights off, use the server to view the M112, dfc signal is in lower case and dw signal in lower case 3.ERR fault lamp is often bright, then view the M23 fault code, if ERR fault lamp is not bright, you can view the M23 fault code or M121 logic fault 4.Check M1311TOP value
4	Elevator has emergency stop in reset operation, not arrive at the terminal station and can't continue operation	1, safety signal is interrupted in operation; 2, door lock signal is interrupted in operation; 3, system has fault in operation; 4, photoelectric switch fault in operation, with output all the time, and stop after encountering strong reduction signal (1/2LS) ; 5, location signal is not normal in operation (1/2LS) , with output all the time, and stops after encountering photoelectric signal;	1. ES on the motherboard lights up, use the server to view M112, ES signal in capital means safety circuit is disconnect, check the safety switch action; 2. DFC or DW on the motherboard lights off, use the server to view the M112, dfc signal in lower case means the landing door lock disconnected, and dw signal in lower case means car door lock disconnected 3.ERR fault lamp is often bright, then view the M23 fault code, if ERR fault lamp is not bright, you can view the M23 fault code or M121 logic fault 4. If M1121/2LV signal shows uppercase whether in leveling or non-leveling position, it means photoelectric switch fault 5. if M1121/2LS signal shows uppercase whether in strong reduction position or non-reduction position, it means 1/2LS is not normal
5	Elevator overrunning	1, strong reduction (1/2LS) signal fault; 2, photoelectric signal fault no output; 3, start sliding distance too big; 4, stop sliding distance too big; 5, UIS/DIS position is inverse, overrunning in re-leveling operation	1.Run at slow speed to confirm the M112 strong reduction (1/2LS) signal is in uppercase when the elevator does not enter the position of actual strong reduction action, if itis always in lowercase, it indicates strong reduction (1/2LS) signal fault 2.When elevator runs to the bottom floor/top floor leveling, M1121/2LV signal will not change to uppercase 3.Check whether the sliding distance will cause upper limit switch action when elevator starts at the top 4.Check whether the sliding distance will cause lower limit switch action when elevator stops at bottom floor

			5.Run at slow speed and check whether M112UIS/DIS signal action sequence is correct
6	Elevator collapsing to the bottom of bit	1, strong reduction signal fault; 2, photoelectric signal fault has no output; 3, start sliding distance too big; 4, stop sliding distance too big; 5, UIS/DIS position is reverse, collapse to the bottom of bit in re-leveling operation;	1. Run at slow speed to confirm the M112 strong reduction (1/2LS) signal is in uppercase when the elevator does not enter the position of actual strong reduction action, if it is always in lowercase, it indicates strong reduction (1/2LS) signal fault 2.When elevator runs to the bottom floor/top floor leveling, confirm whether M112 1/2LV signal is in uppercase 3.Check whether the sliding distance will cause upper and lower limit switch action when elevator starts at the top/bottom floor; 4.Check whether the sliding distance will cause upper and lower limit switch action when elevator stops at top/bottom floor 5.Run at slow speed and check whether M112 UIS/DIS signal action sequence is correct 6.Monitor M211 Now position m, and confirm the pulse counting direction
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
7	When elevator is in maintenance start time, main engine has abnormal sound, ERR light up	Drive has fault;	View M23 fault code
8	Elevator overruns in maintenance start time, and can't run again	1, Drive has fault; 2, Encoder line broken;	1. View M23 fault code 2.Use a universal meter to measure the connection and disconnection of each wire at the ends of encoder

9	Press ascending and descending button, the elevator unable to move	1, No safety signal; 2, No door lock signal; 3, No door closing in-place signal; 4, Enable parameter setting error; 5, System has fault;	1.Use the server to view M112, ES signal in capital means safety circuit is disconnected, check the safety switch action; 2.Use the server to view M112, dfc signal in lowercase means landing door lock disconnected and dw signal in lowercase means the car door lock disconnected 3. If M112 DFC and DW are both in uppercase, dcl door closing in-place signal in lowercase 4.M222RUN source SVT shall be 0 5. View M23 fault code
10	Press ascending maintenance button, the elevator runs upward, and press descending maintenance button, the elevator not move	1, control panel not receive the instructions of ascending operation; 2, there is 1LS/ photoelectric signal at the same time;	1.press maintenance descending button, M11 dib not change to be uppercase 2. view M1121LS in uppercase, and 1LV in uppercase
11	Press descending maintenance button, the elevator runs downward, and press ascending maintenance button, the elevator not move	1, control panel not receiving instructions of ascending; 2, there is 2LS/ photoelectric signal at the same time;	1.maintenance ascending button, M22 uib not change to be uppercase 2. view M112 2LS in uppercase, and 2LV in uppercase
12	Press descending maintenance button, the elevator runs downward, and press ascending maintenance button, the elevator stops	1, drive fault; 2, braking resistance fault, causing DC but overvoltage fault;	1. view M23 fault code 2. remove the brake resistance line, use a universal meter to measure resistance value, if the resistance is damaged, the measured resistance value and the standard resistance on the braking resistor box will have a great deviation

	after running for some distance		
13	Press ascending maintenance button, the elevator runs upward, and press descending maintenance button, the elevator stops after running for some distance	1, drive fault; 2, braking resistance fault, causing DC but overvoltage fault;	1. view M23 fault code 2. remove the brake resistance line, use a universal meter to measure resistance value, if the resistance is damaged, the measured resistance value and the standard resistance on the braking resistor box will have a great deviation
14	Elevator can only be in maintenance, and can't have well self learning operation	1, safety circuit disconnected in normal status; 2, Run enable not set as 0; 3, system is in maintenance mode;	1.M112ES signal is in uppercase and, dfc or dw lowercase 2 confirm M222Run enable =1 3.M11 status displays INS, and M112ERO is in uppercase
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
15	After the elevator self learning is completed, can't use the server to call the elevator	1, Run enable not set as 0; 2, No DCS operation, or not cancel DCS operation; 3, EN-CRT=0, can't run above 16floors; 4, System not in IDLE state;	1.confirm M222Run enable =0 2.M111 elevator current state will display Start DCS 3.when floor is greater than 16m M1310EN-CRT=0 4.M111 elevator status is not IDLE
16	Press maintenance button, elevator continues to slide towards the heavy side	1, pre torque compensation is not suitable;; 2, track err function is shielded;	1.Confirm M224 KP and KI value, make adjustment 2.M224Track error is set as 0

17	Normal running, elevator slides 2~4cm towards heavy side, and then runs to the designated direction	1, pre torque compensation is not suitable;; 2, driver board J11/J12 short circuit cap is short-circuited to the right or J11/J12 not installed;	1 Confirm M224 KP and KI value, make adjustment 2. driver board J11/J12 short circuit cap is short-circuited to the right or J11/J12 not installed
18	Elevator takes correction running after arriving at target floor	system has fault;	View M23 fault code or view M121 logic fault
19	Elevator makes emergency stop in fast running process, and then makes correction running	1, safety signal is interrupted in running process; 2, door lock signal is interrupted in running process; 3, system has fault in running process;	1.ES lights up on the motherboard, use the server to view the M112, ES signal in uppercase means safety loop disconnected, check the safety switch action 2. DFC or DW lights off on the motherboard, use the server to view the M112, dfc signal in lowercase means the landing door lock is off, dw signal in lowercase means the car door lock off 3.View M23 fault records or M121 logic fault
20	Elevator runs upwards for reset	1. upper strong reduction switch has no action or installation distance too short 2. inserting plate on the top floor is wrong in position 3. photoelectric failure	1.2ls of M112 has been in lowercase without action or the position of action is too short from the leveling of the top floor (850mm distance for 1m/s, 1750mm for 1.5m/s, and 2200mm for 1.75m/s) 2.inserting plate installation position on the top floor is too close to the upper limit location (150mm is the standard) 3.When it runs to another floor the 1lv and 2lv of M112 will not turn to uppercase
21	Elevator collapse to the bottom of bit for reset	1. lower strong reduction switch has no action or installation distance too short 2. inserting plate on bottom floor 3.photoelectric failure	1. M112 has been in lowercase without action or the position of action is too short from the leveling of the bottom floor (850mm distance for 1m/s, 1750mm for 1.5m/s, and 2200mm for 1.75m/s) 2.inserting plate installation position on the bottom floor is too close to the lower limit location (150mm is the standard) 3. When it runs to another floor the 1lv and 2lv of M112 will not turn to uppercase

Serial No.	Fault description	Possible cause	Confirmation method
22	When the elevator runs to the target floor, it is found higher than the leveling	1. inserting plate installation position not accurate; 2.photoelectric distance not accurate;	1.run at slow speed to adjust the inserting plate position of each floor; 2.adjust upper and lower photoelectric distance and UP level and DOWN level value of M223
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
23	When the elevator runs to the target floor, it is found lower than the leveling	1. inserting plate installation position not accurate; 2.photoelectric distance not accurate	1. run at slow speed to adjust the inserting plate position of each floor; 2.adjust upper and lower photoelectric distance
24	When the elevator starts, it has high frequency jitter	1, pre-torque adjustment not accurate; 2, main engine positioning not accurate;	1.confirm KP and KI value of M224 2. check M225 main engine parameters and M226Drive size
25	When elevator starts, main engine has big noise	1, main engine parameter and frequency converter parameter setting is not suitable, inertia, inductance, resistance, carrier; 2, torque compensation parameter KP and KI not suitable; 3, bake not completely opened;	1.check M225 main engine parameters and M226Drive size, confirm M224Inertia kg/m2, M225Ld0, Lq0, Ld, Lq, and M226Switch frequency 2.confirm KP and KI value of M224 3.confirm brake gap is too small
26	When elevator runs, main engine has big noise	1, inverter parameter setting is not appropriate, such as inertia and inductance setting is too large; 2, carrier frequency setting is unreasonable;	Check M225 main engine parameters and M226Drive size, confirm M224Inertia kg/m2, M225Ld0, Lq0, Ld, Lq, and M226Switch frequency
27	Too big noise when elevator stops	1, inverter parameter setting is not appropriate, such as inertia and inductance setting is too large; 2, carrier frequency setting is unreasonable;	Check M225 main engine parameters and M226Drive size, confirm M224Inertia kg/m2, M225Ld0, Lq0, Ld, Lq, and M226Switch frequency

28	Elevator runs at half speed	1, no self-learning running; 2, parameter setting error; (rope winding ratio and traction wheel diameter error, Velocity normal error)	1.check whether the height of each floor and the stroke switch in M214 is consistent with the field size; 2.check whether the rope winding ratio in M222,traction wheel diameter and M222Velocity normal is consistent with the field size
Door fault			
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
1	Door is not opened when elevator runs to the target floor, and door is opened on the next floor, unable to run after trying three times	1, door machine is not normal, unable to completely open the door; 2, DOL signal has been there;	1.after door machine receives DO signal, the door gets stuck and can't be opened or can't be completely opened 2.in M112, DOL has been in uppercase when the door is not opened in place
2	Door is not opened when elevator runs to the target floor	1.DDO status; 2.door machine type in control panel is not correctly set; 3. after reaching the station, the system has fault and door can't be opened; 4. door machine wiring error; 5.door machine parameter setting error, not accept the system instructions;	1.M111 door status displays DDO 2.M1315DOOR value, relay control mode door machine is set as 5, DT code is set as 14 3.M23 to view drive fault code and, M121to view logic fault code 4. check door opening signal wiring 5.check door machine parameter
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
3	Keep door opening and closing action when elevator runs to the target floor	1, screen signal repeats action; 2, screen, light eyes, safety plate address setting error, resulting in the continuous existence of signal	1.in M112LRD repeats uppercase and lowercase 2. in M132 check EDP,SGS and LRD address
4	Door keeps opening without closing	1, screen, light eyes, and safety plate signal has been always there;	1.DOS,LRD,EDP in M112 has been in uppercase 2.LWO in M112 has been in uppercase

	when elevator runs to the target floor, buzzer output	2, overload signal has been always there;	
5	Elevator door is not opened in place, always in half-opening state	1, door machine gets stuck and unable for action; 2, door machine conversion board is not normal;; 3, LDR parameter setting error; 4, enter maintenance mode in door opening and closing process;	1.check whether the door is stuck by a foreign body; 2.check whether the door opening and closing signal on conversion board, door opening and closing in place signal, and screen signal action is normal or not; 3.M1315LDR=1 4.in elevator door opening and closing process, check whether M112ERO has changed to uppercase
6	Elevator door is not closed in place, always in half opening state	1, door machine loses power; 2, door machine gets stuck; 3, enter maintenance mode in door opening and closing process	1.go to the car top to check whether door machine loses power 2.check whether the landing door on the problem floor is stuck by foreign body 3. in elevator door opening and closing process, check whether M112ERO has changed to uppercase
7	Close the door with too much force and landing door collides	Door machine parameter setting error	check door machine parameter
8	Door is open and elevator can run	door lock is short circuited	In door opening state, M112DFC and DW in uppercase

Internal and external call fault

Serial No.	Fault description	Possible cause	Confirmation method
1	Control box button is pressed, but there is a response.	1, button lamp is broken; 2, button lamp circuit wiring problem; 3, RS5 broken;	1. replace with a good button, problem solved 2.replace with a good button, problem not solved 3.replace RS5 board

2	Control box button is pressed, but there is no light or response.	1, communication failure, RSL and the motherboard RS5 communications are faulty; 2, address setting error;; 3, floor response enable setting error; 4, if there is only one button that does not respond, the button may be damaged or the control box wiring has problem	1. check RSL communication wiring 2.M132 check the floor button address 3.M1331 check the floor response enable setting 4.replace with a good button
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**
3	Press the control box button bright, but no response	1, elevator has fault on other floor, for example, some person or object blocks the screen for long; 2, door machine has fault, unable to close the door in place or no door lock signal, and elevator can't run;	1. check where the door on other floor is blocked by person or object 2.dcl in M112 is in lowercase and dfc or dw in lowercase
4	Elevator runs to wrong floor	1, check whether there is any blind floor; 2, I/O address setting error;	1.confirm whether M1331 has virtual floor setting 2. check whether M132IO address and IO list in schematic diagram is consistent
5	Button flashes when elevator is running	1, terminal absorption plate configuration error; 2, RSL signal is affected by strong current interference;; 3, I/O address setting error;	1.check whether the terminal absorption plate is missing or burned out; 2 confirm RSL signal line is isolated from strong current; 3.check IO address of M132 with the address table on schematic diagram 4.checkJ8, J10, J15, J16 jumper on motherboard, parallel shared call or group control two shared call shall cancel the 4 jumpers.
Display fault			
Serial No.	**Fault description**	**Possible cause**	**Confirmation method**

Attachment

1	The monitor displays the floor incorrectly	1, display setting error; 2, there is blind floor but blind floor display is not set;	Check M134 floor display setting
2	The monitor does not display the image, but still can normally call elevator	1, plug-in (H) is not inserted or in poor contact; 2, monitor is not good;	1.check the plug-in (H) on the monitor 2.replace with a good monitor and display normally
3	Monitor has splash screen	monitor is not good	The display is normal after a good monitor is installed
4	Monitor has code missing phenomenon	Monitor is not good	The display is normal after a good monitor is installed
5	Monitor has been running fully loaded	1, full-load signal has been input; 2, monitor is not good; 3, full-load address is set as other I/O	1.whether LNS address in M132 is mandatorily input 2.replace the monitor 3. whether LNSL address in M132 is duplicate with other IO settings
6	Monitor has been displaying fire control	1, fire control signal has been input; 2, monitor is not good 3, fire lamp address is set as other I/O	1.whether EFK address in M132 is mandatorily input 2. replace the monitor 3.whether FSL address in M132 is duplicate with other IO settings
7	Monitor has been displaying overload	1, overload signal has been input; 2, monitor is not good; 3, overload lamp address is set as other I/O	1. whether LWS address in M132 is mandatorily input 2. replace the monitor 3. whether OLS address in M132 is duplicate with other IO settings
8	Floor information displayed and actual floor	1, acceleration and deceleration setting in the drive is set too small;	Accelera normal and Decelera normal value in M222 is set too small